

Do daily habits predict obesity?

A data minning perspective

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Overview



01 Introduction

02 Literature Review

03 Data Description

04 Methodology

05 Univariate Analysis

06 Bivariate Analysis

07 ANOVA - test

08 Linear Regression

09 PCA

10 Conclusion

Introduction

- Behavioral, genetic, and environmental factors all contribute to obesity, which is a serious problem for public health. The frequency of eating, physical activity, alcohol consumption, snacking, and mode of transportation are the primary focus of this study.
- **Research Question :** Can we identify which daily behaviors are statistically associated with different obesity levels?



sector

According to WHO (World Health Organization), poor diet and lack of movement are the main causes of obesity.



Literature Review

Popkin et al. (2002) and Swinburn et al. (2011) explained that modern life makes people more inactive and exposed to unhealthy food. These factors increase obesity worldwide.



Other studies found that watching too much TV, using computers for many hours, and drinking sugary drinks are also linked to weight gain.



Dataset and Variables

Number of observations : 2111 Observation

Number of columns : 17 column

Variables :

Numeric Variables :

- Age in years
- Height in meters
- Weight in Kg
- CH2O (Daily water intake): 1–3

Categorical Variables :

- Gender : Male or Female
- family_history_with_overweight: Yes or No
- FAVC (Frequent consumption of high-calorie food): Yes or No
- FCVC (Frequency of vegetable consumption): 1–3
- NCP (Number of main meals): 1–4+

- CAEC (Eating between meals): No, Sometimes, Frequently, Always
- SMOKE: Yes or No
- SCC (Calories monitoring): Yes or No
- FAF (Physical activity frequency): 0–3
- TUE (Time using tech devices daily): 0–2
- CALC (Alcohol consumption): No, Sometimes, Frequently, Always
- MTRANS (Transport used): Public, Bike, Motorbike, Walking, etc.
- Obesity level (Obesity level):
Underweight, Normal, Overweight, Obesity Type I–III

Methodology

Tools Used : R language and Rstudio for all statical calculation and some graphs

1- Univariate Analysis :

1.1 Empirical Analysis for numeric variables

- Summary statistics such as mean, median, standard deviation, minimum, maximum, and quantiles are calculated for Age, Weight, and Height to understand their central tendency and variability.

1.2 Empirical Analysis for categorical variables

- The frequency of each category is calculated for variables such as Gender, Obesity level, CAEC, family_history_with_overweight, CALC and FAF.

1.3 Charts for categorical variables

- Bar charts are used to visualize the distribution of categories in variables such as Gender, Obesity level, CAEC, FAVC, CALC while Pie charts are used to visualize the family_history_with_overweight.

1.4 Charts for numerical variables

- Histograms with density curves are used to visualize the distribution and variability in Age, Weight, and Height.

Methodology

2- Bivariate Analysis :

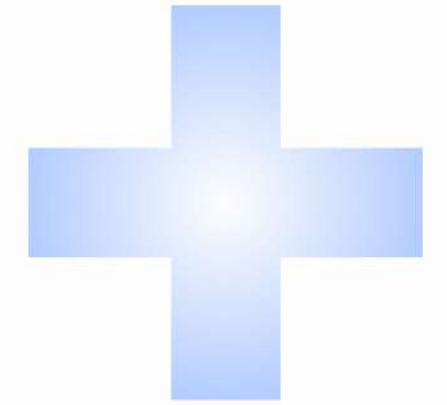
2.1 Visualization of Categorical versus Numeric Variable

- Column Chart: Used to compare the mean values of the numeric variable across different categories. Box Plot: Helps assess the distribution, spread, and presence of outliers within each category.

2.2 Empirical Analysis

- ANOVA Test (Analysis of Variance): Conducted to determine if there is a statistically significant difference in the means of the numeric variable across different categories of the categorical variable.

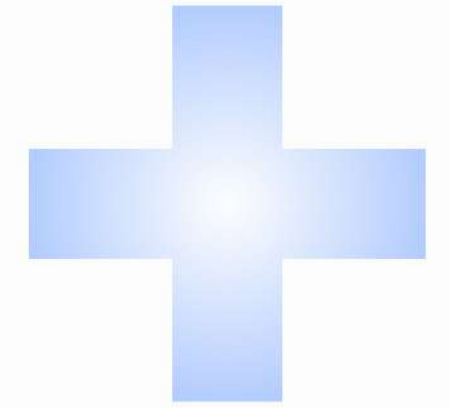
Univariate Analysis



5.1 Numeric Variables

Variable	Min	1st Quartile	Median	Mean	3rd Quartile	Max	Standard deviation
Age	14.00	19.95	22.78	24.31	26.00	61.00	6.34
Height (m)	1.450	1.630	1.700	1.702	1.768	1.980	0.093305
Weight (kg)	39.00	65.47	83.00	86.59	107.43	173.00	26.19
CH2O (L)	1.00	1.58	2.00	2.00	2.47	3.00	0.61

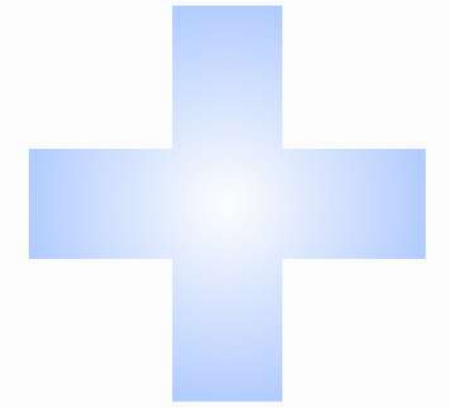
Univariate Analysis



5.2 Categorical Variables

Category	Count
Gender:	
Male	1068
Female	1043
Family History	
Yes	1726
No	385
FAVC (Frequent consumption of high-calorie food)	
Yes	1866
No	245
CAEC (Eating between meals)	
Sometimes	1765
Frequently	242
Always	53
No	51

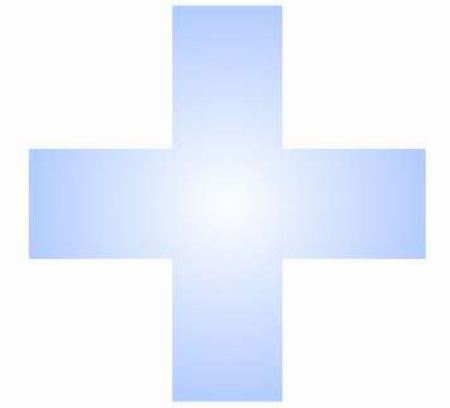
Univariate Analysis



5.2 Categorical Variables

SMOKE	
No	2067
Yes	44
SCC (Calories monitoring)	
No	2015
Yes	96
CALC (Alcohol consumption)	
Sometimes	1401
No	639
Frequently	70
Always	1

Univariate Analysis



5.2 Categorical Variables

MTRANS (Transport used)	
Public Transport	1580
Automobile	457
Walking	56
Motorbike	11
Bike	7
Obesity	
Obesity Type 1	351
Obesity Type 3	324
Obesity Type 2	297
Overweight Level 1	290
Overweight Level 2	290
Normal Weight	287
Insufficient Weight	272

Univariate Analysis

5.3 charts : Categorical Variables

 Bar chart : Gender



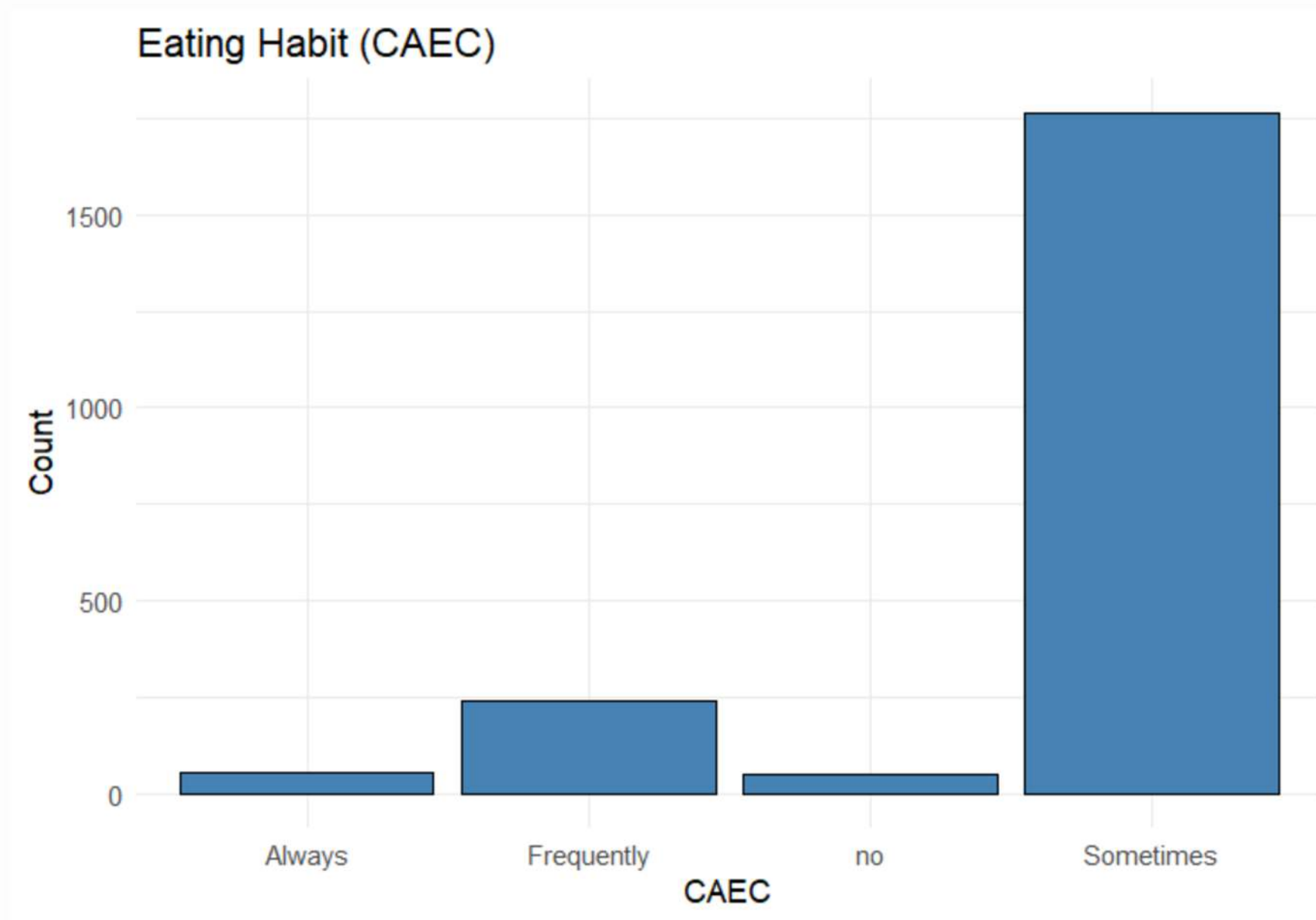
1068 Male
1043 Female

Interpretation: The chart shows the gender distribution of the dataset. There's a nearly equal number of male and female participants. Each group is close to 1,000, suggesting that the dataset is balanced in terms of gender, which is great for avoiding bias in gender-based analysis.

Univariate Analysis

5.3 charts : Categorical Variables

Bar chart : CAEC (Eating between meals)



Sometimes :1765

frequently :242

Always :53

No :51

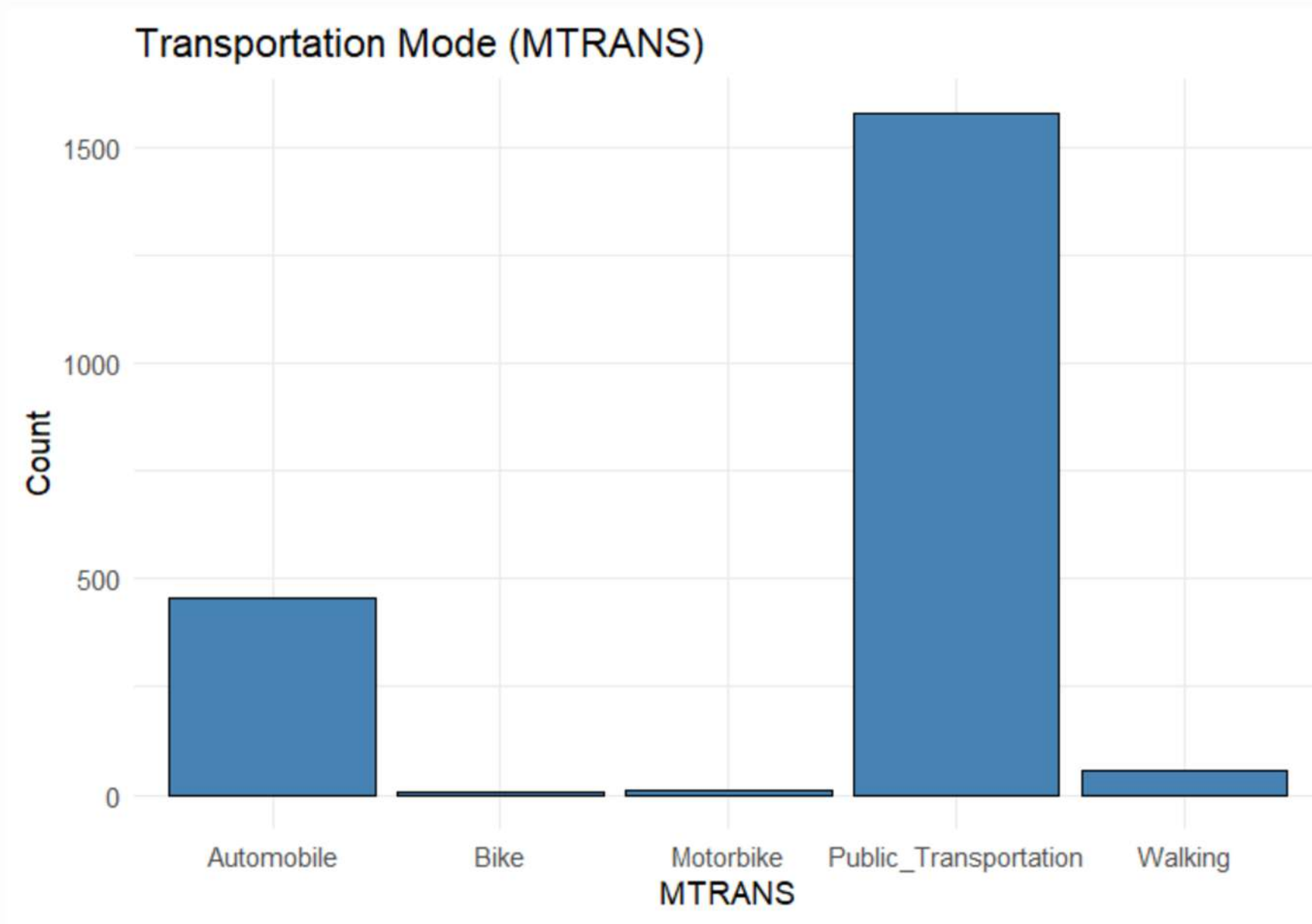
- Snacking is common, but most do it occasionally, rather than habitually.

Interpretation: This bar chart illustrates how frequently people in the dataset eat between meals (CAEC). The vast majority of participants selected "Sometimes", followed by a much smaller number selecting "Frequently", "Always", and "No". This suggests that snacking is common, but most do it occasionally, rather than habitually.

Univariate Analysis

5.3 charts : Categorical Variables

 Bar chart : MTRANS (transport used)



Public Transport :1580

Automobile :457

Walking :56

Motorbike :11

Bike :7

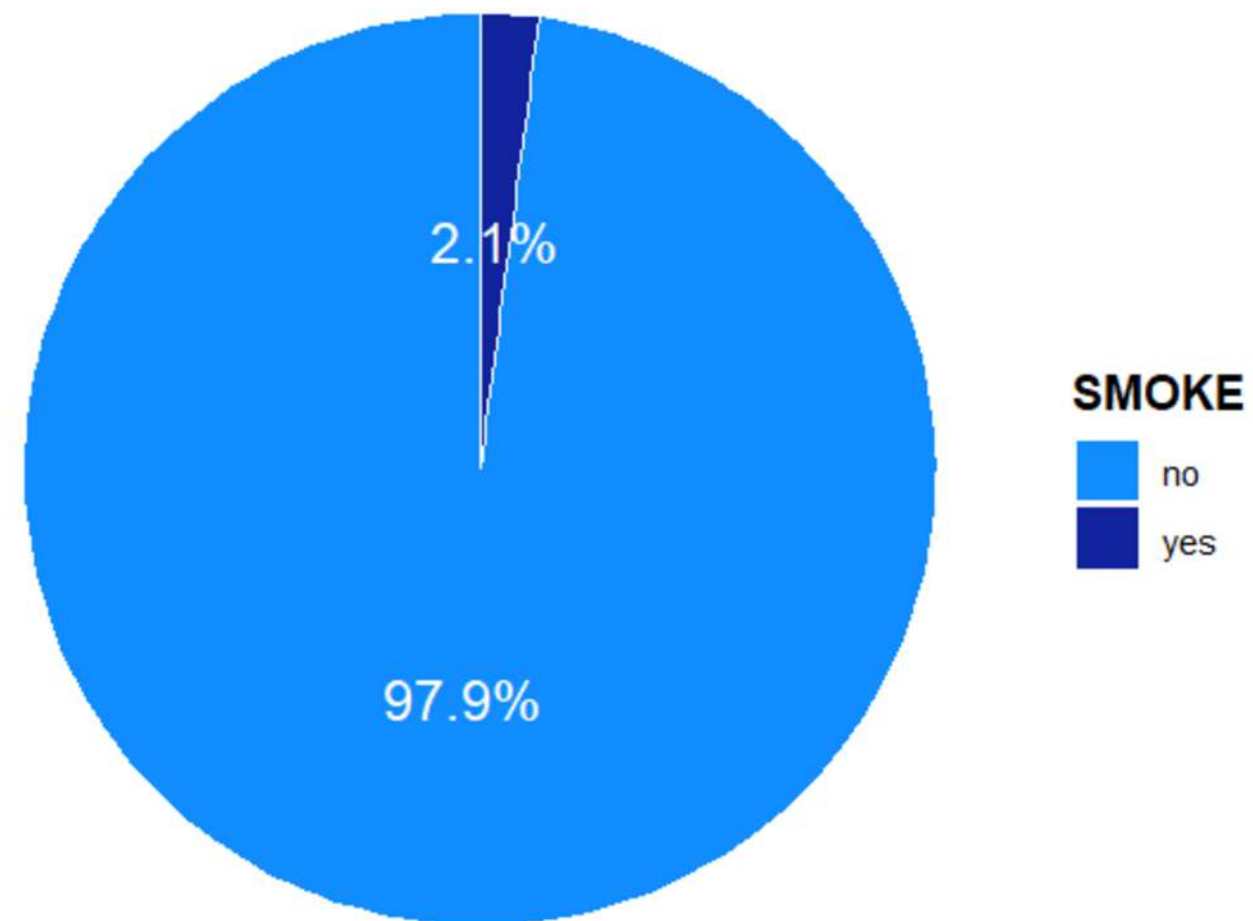
Interpretation: This bar chart shows the modes of transportation used by individuals in the dataset. A large majority use public transportation, followed by a smaller portion who travel by automobile. Very few participants report walking, using a motorbike, or biking as their main means of transportation.

Univariate Analysis

5.3 charts : Categorical Variables

 Pie chart : Smoking

Pie chart of Smoking values



No :2067

Yes :44

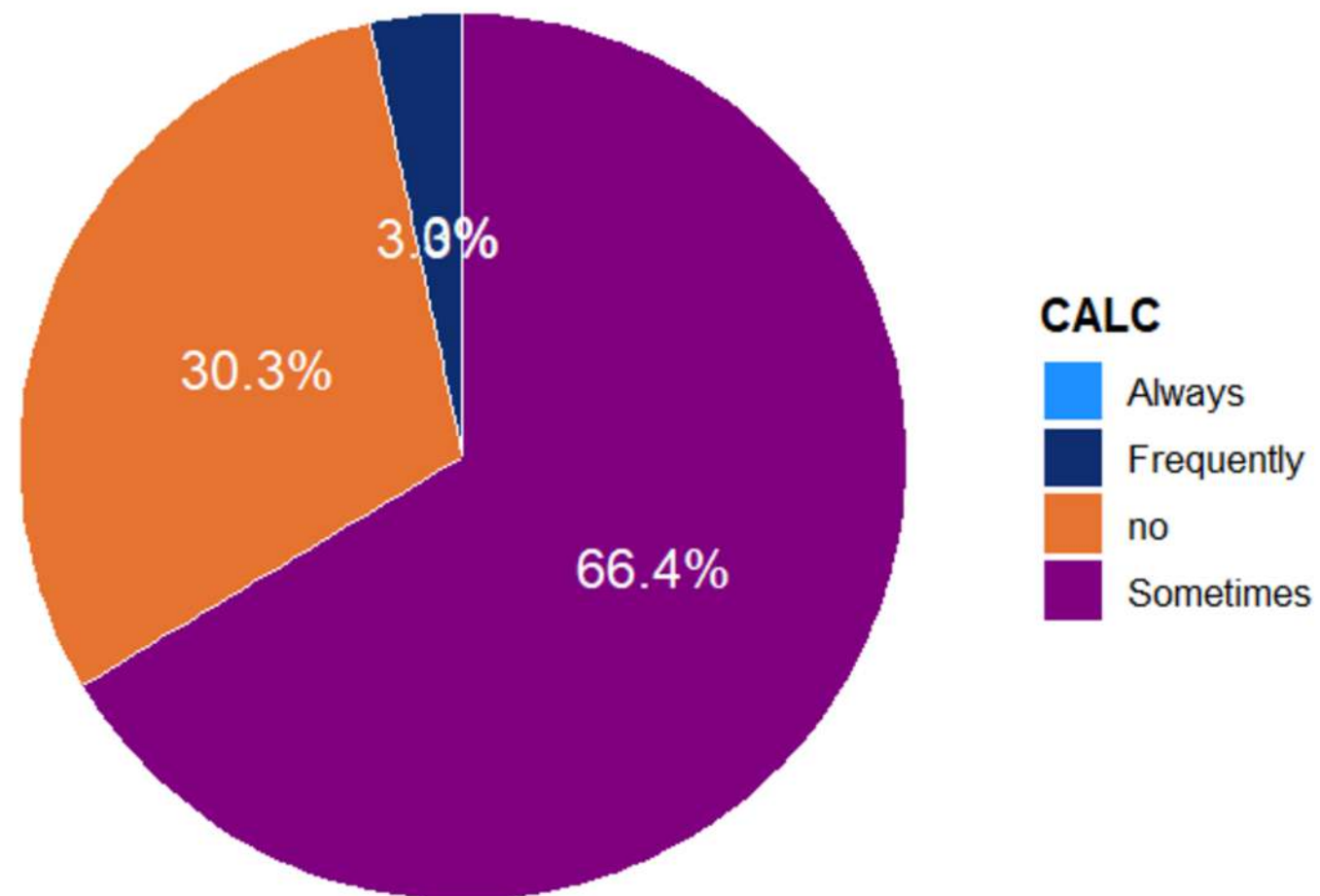
- Smoking is not a common habit among this population.

Interpretation: This pie chart displays the smoking behavior of individuals in the dataset. A very large majority (97.92%) do not smoke, while only a small fraction (2.08%) report that they do. This suggests that smoking is not a common habit among this population.

Univariate Analysis

5.3 charts : Categorical Variables

 Pie chart : CALC (Alcohol consumption)



Sometimes : 1401

No : 639

Frequently : 70

Always : 1

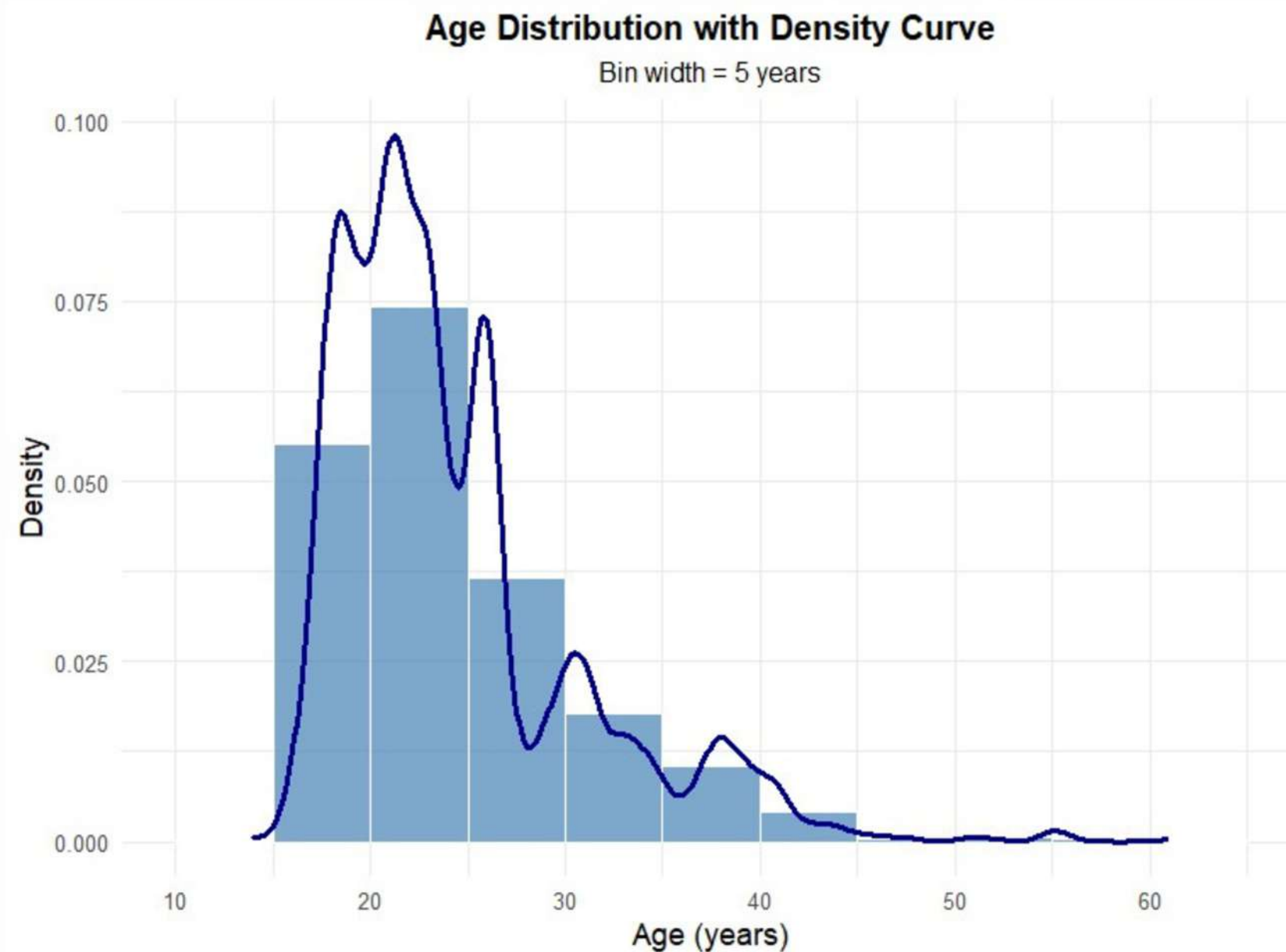
- Smoking is not a common habit among this population.

Interpretation: This pie chart illustrates the frequency of alcohol consumption among individuals in the dataset. The majority (66.73%) consume alcohol "Sometimes", followed by 30.27% who report "No" alcohol use. Only a small fraction report drinking "Frequently" (3.32%) or "Always" (0.32%). This suggests that occasional alcohol consumption is the norm, with few regular drinkers.

Univariate Analysis

5.4 charts : Numerical Variables

Histogram and density curve : Age



Interpretation : the majority of individuals in the dataset are between 15 and 30 years old, with noticeable peaks around ages 18–22 and 25. After age 30, the distribution drops off significantly, with very few individuals above age 40.

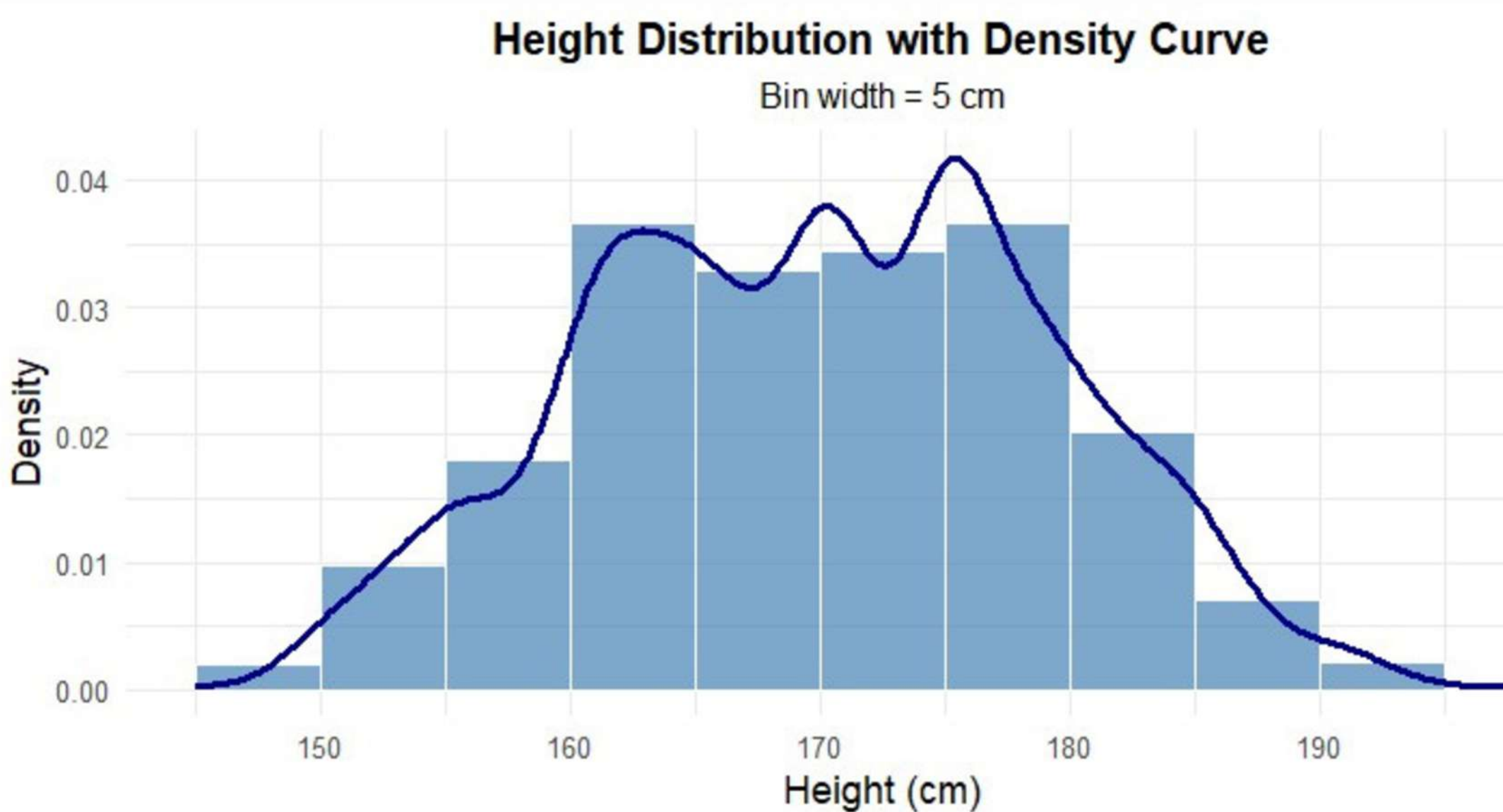
- the population is predominantly young adults, likely students or early-career individuals.

Economic Interpretation: A younger population like this often represents individuals who are in transitional economic stages ; either pursuing education, entering the job market, or just beginning to establish financial independence. At this age, lifestyle choices (such as diet, exercise, and habits like smoking or alcohol use) can have long-term consequences on both health outcomes and future economic productivity.

Univariate Analysis

5.4 charts : Numerical Variables

Histogram and density curve : Height (cm)



Interpretation : The visualization shows a roughly symmetrical distribution centered around 165–175 cm, with a slight bimodal tendency. Most individuals fall within the 160–180 cm range, with fewer individuals at the extremes (below 150 cm or above 190 cm).

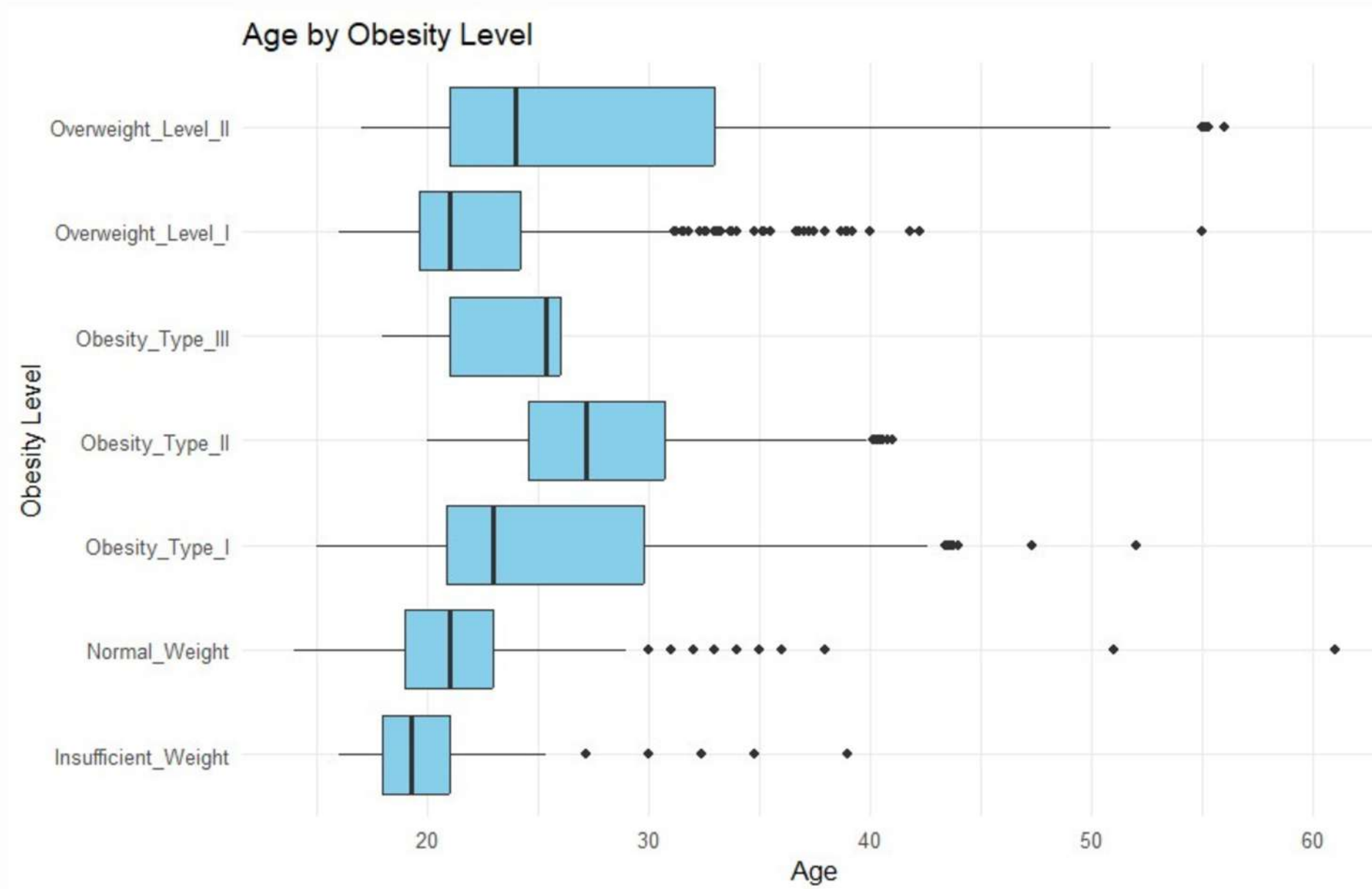
- This suggests a fairly balanced representation of average adult height.

Economic Interpretation: Height has been historically correlated with certain socioeconomic factors. In many cases, taller individuals benefit from early-life access to better nutrition, healthcare, and living conditions, which contribute to growth. Thus, this distribution may reflect inequalities in childhood development, influenced by family income and education levels.

Bivariate Analysis

6 Visualization of Categorical vs Numerical Variables

 Boxplot : Age by Obesity Level



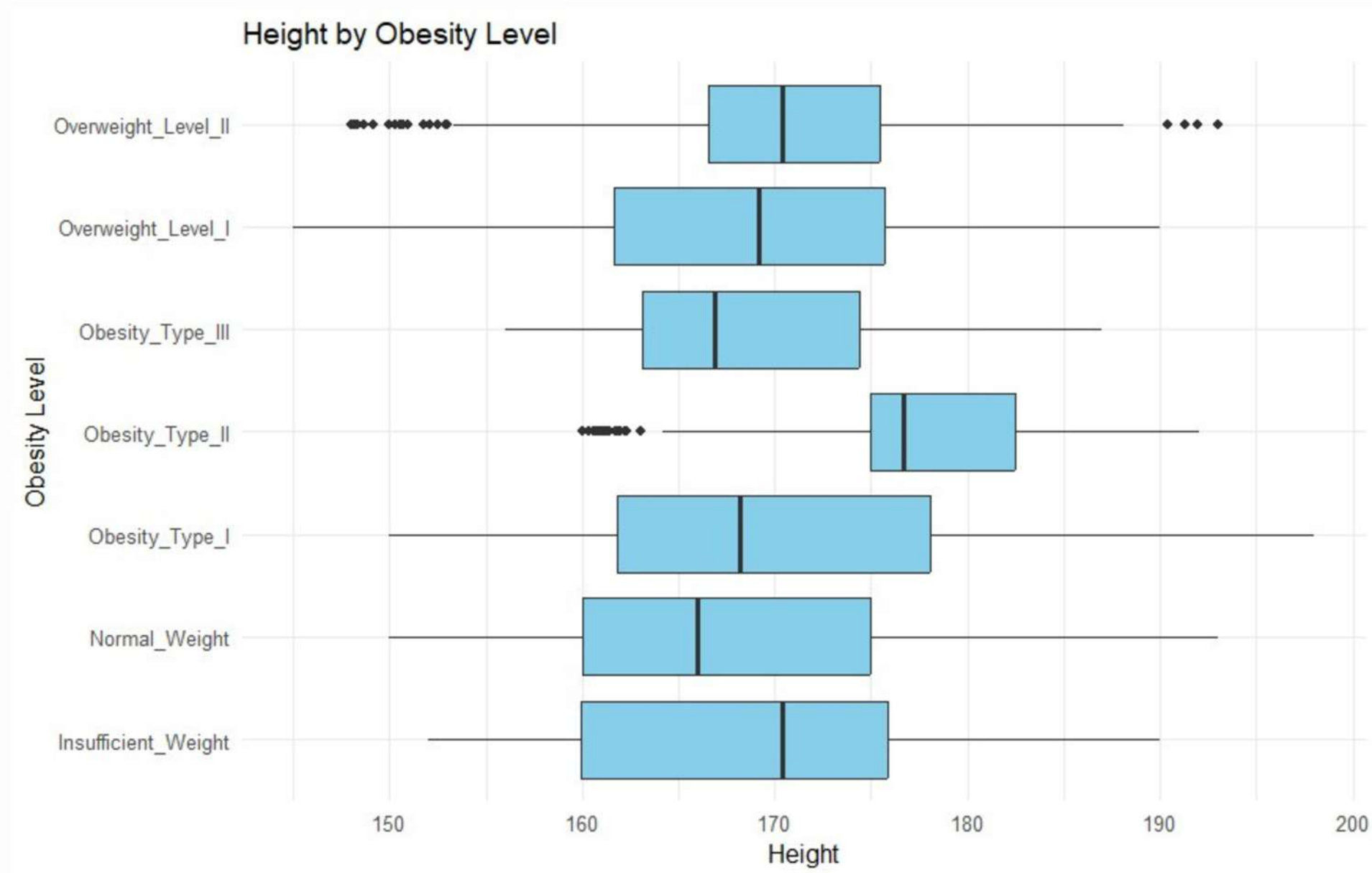
Interpretation :

- The median age for individuals with Obesity_Type_III and Obesity_Type_II is higher compared to those with Normal_Weight or Insufficient_Weight, suggesting that older age may be associated with higher obesity levels.
- Younger individuals tend to fall into the Normal_Weight or Insufficient_Weight categories, indicating a potential trend where weight issues become more prevalent with age

Bivariate Analysis

6 Visualization of Categorical vs Numerical Variables

 Boxplot : Height by Obesity Level



Interpretation :

- There is no clear trend between height and obesity level, as the median height remains relatively consistent across all categories (Insufficient_Weight to Obesity_Type_III).
- Some outliers may exist in certain groups (in very tall or short individuals as shown), but overall, height does not appear to be a major factor in determining obesity level.

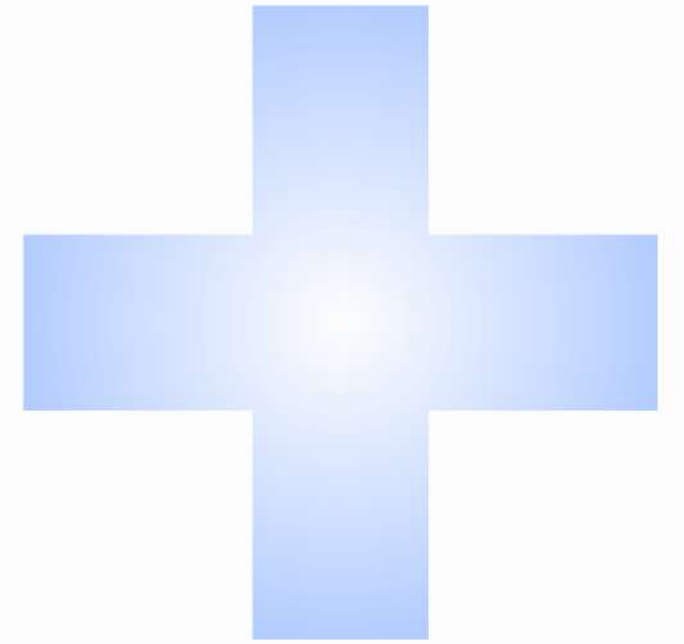
ANOVA – test

7. ANOVA - test

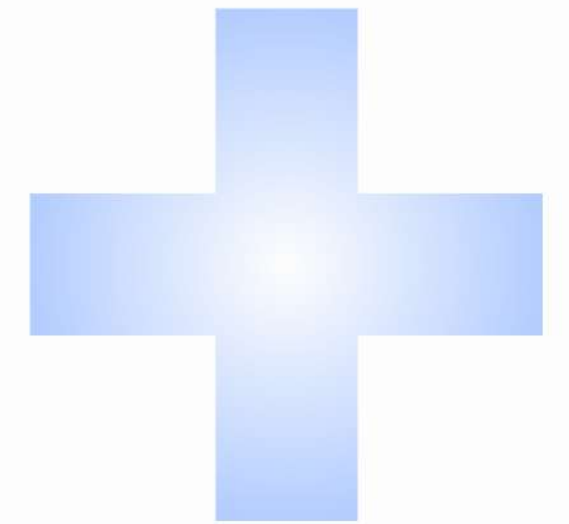
ANOVA (Analysis of Variance) is a method used to compare the means of three or more groups to see if they are significantly different.

- Null hypothesis (H_0): All group means are equal.
- Alternative hypothesis (H_1): At least one group mean is different.

If the p-value < 0.05 , we reject H_0 and say there is a significant difference between the groups.



ANOVA – test



7. ANOVA - test

Obesity Level with Weight

Null hypothesis (H_0): Obesity level has no effect on weight. (The mean weight is the same across all obesity groups.)

Alternative hypothesis (H_1): Obesity level has an effect on weight. (The mean weight is different for at least one obesity group.)

```
> aov_weight <- aov(Weight ~ Obesity, data = df)
> summary(aov_weight)
```

	Df	Sum of Squares (Sum Sq)	Mean Squares (Mean Sq)	5. p-value (Pr(>F))
Obesity	6	1,228,371	204,729	< 2e-16

ANOVA – test



7. ANOVA - test

Obesity Level with Height

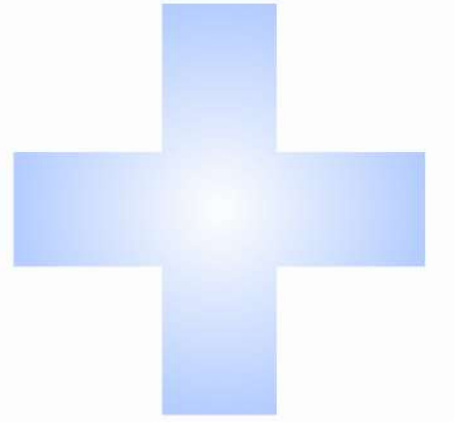
Null Hypothesis (H_0): The mean height is equal across all obesity categories.

Alternative Hypothesis (H_1): At least one obesity category has a significantly different mean height.

```
> aov_weight <- aov(Height ~ Obesity, data = df)
> summary(aov_weight)
```

	Df	Sum of Squares (Sum Sq)	Mean Squares (Mean Sq)	5. p-value (Pr(>F))
Obesity	6	1.814	0.30240	< 2e-16

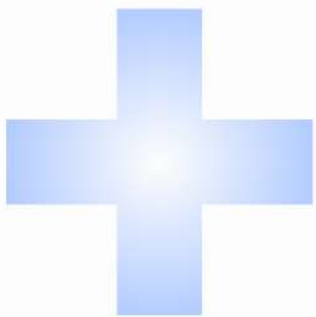
Linear Regression



8. Linear Regression

Linear regression is one of the most common statistical methods used to understand the relationship between two or more variables. It helps us see how changes in one variable (called the independent variable) affect another variable (called the dependent variable).

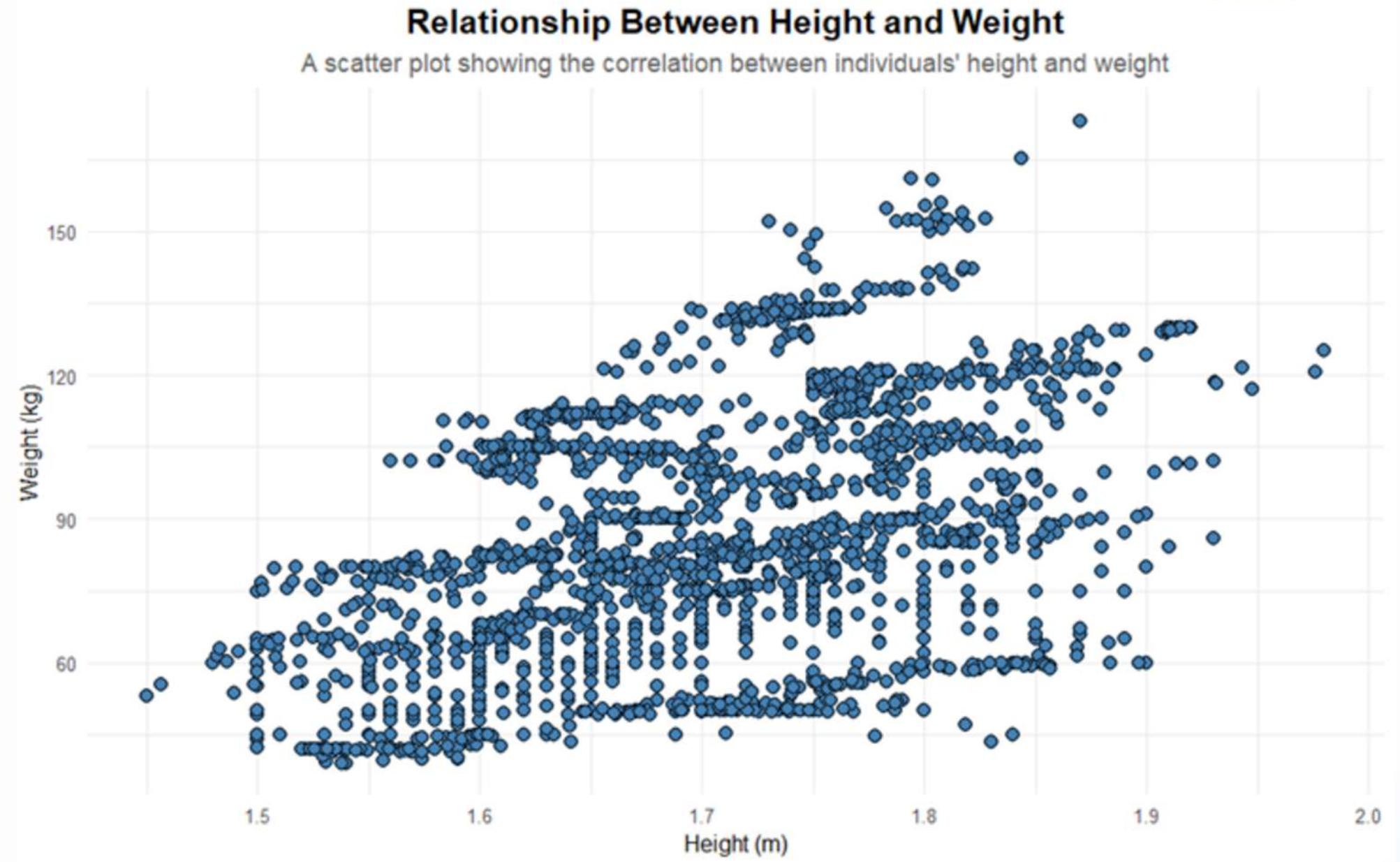
In simple terms, linear regression draws a straight line that best fits the data points. This line can be used to make predictions or to better understand patterns in the data.



Linear Regression

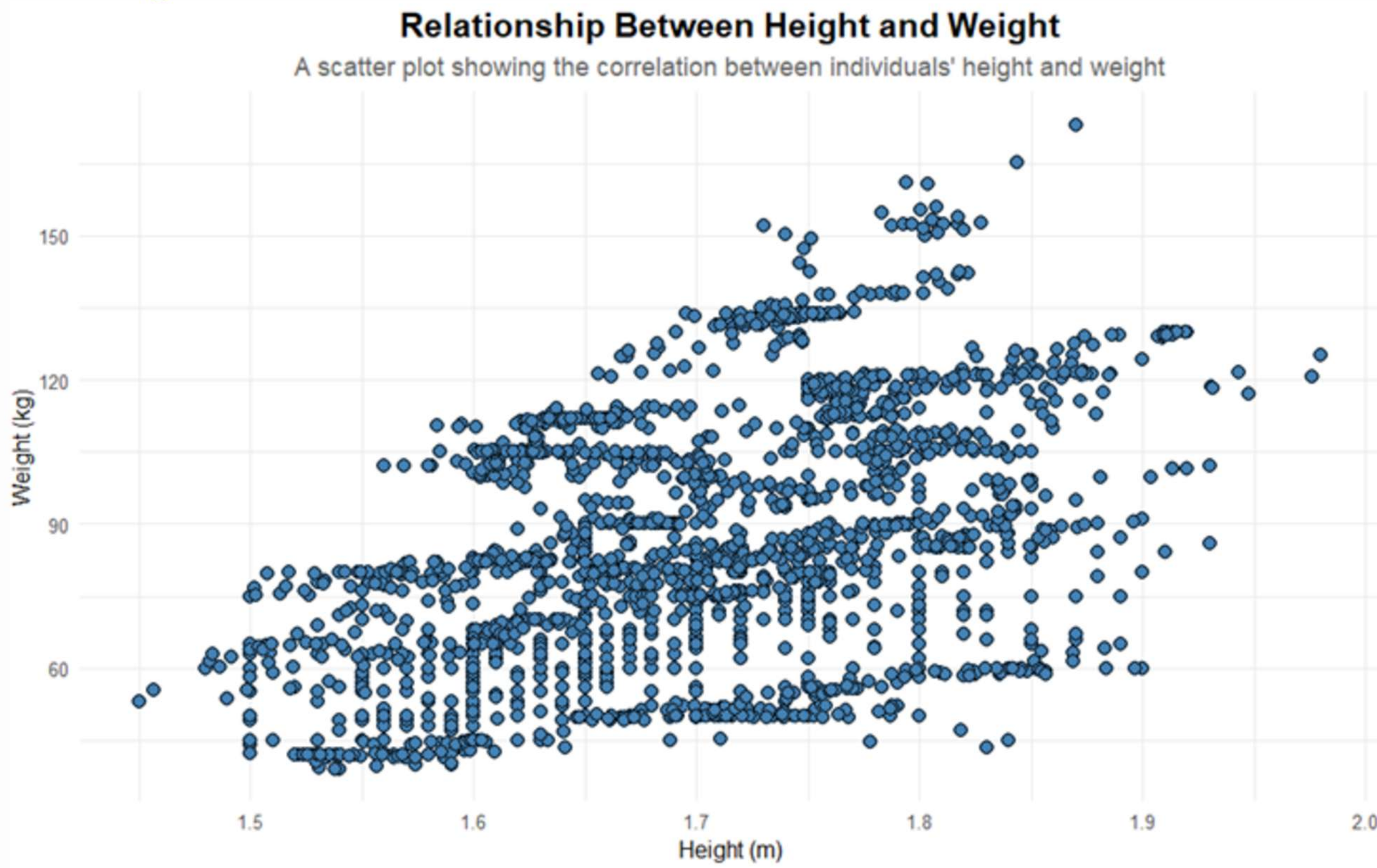
8. Linear Regression

if we plot the height and the weight, we will get a kind of a upwarding trend no one from that that they are positively correlated so if one increases the other more likely to increase as shown in the following plot:



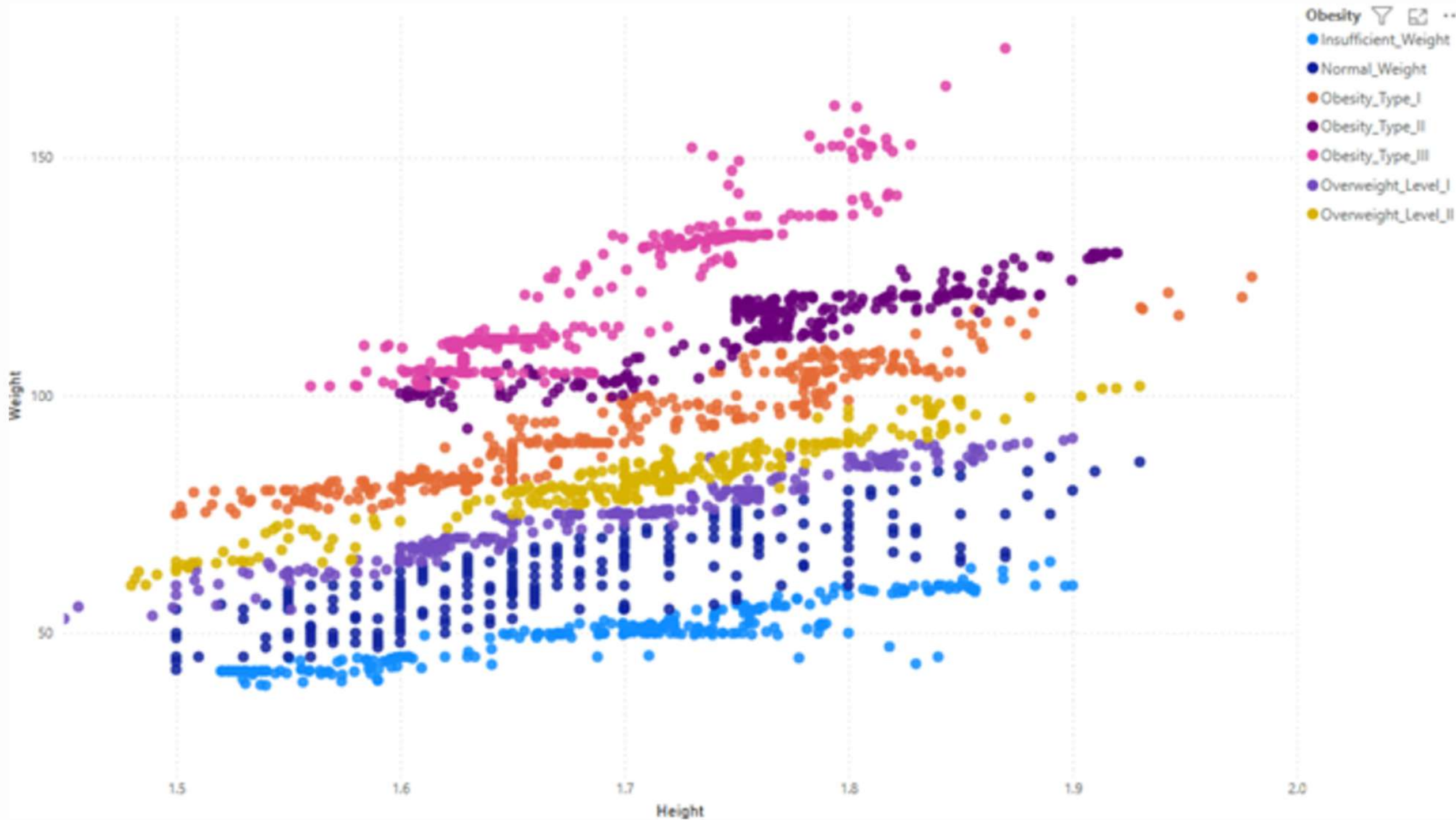
Linear Regression

8. Linear Regression



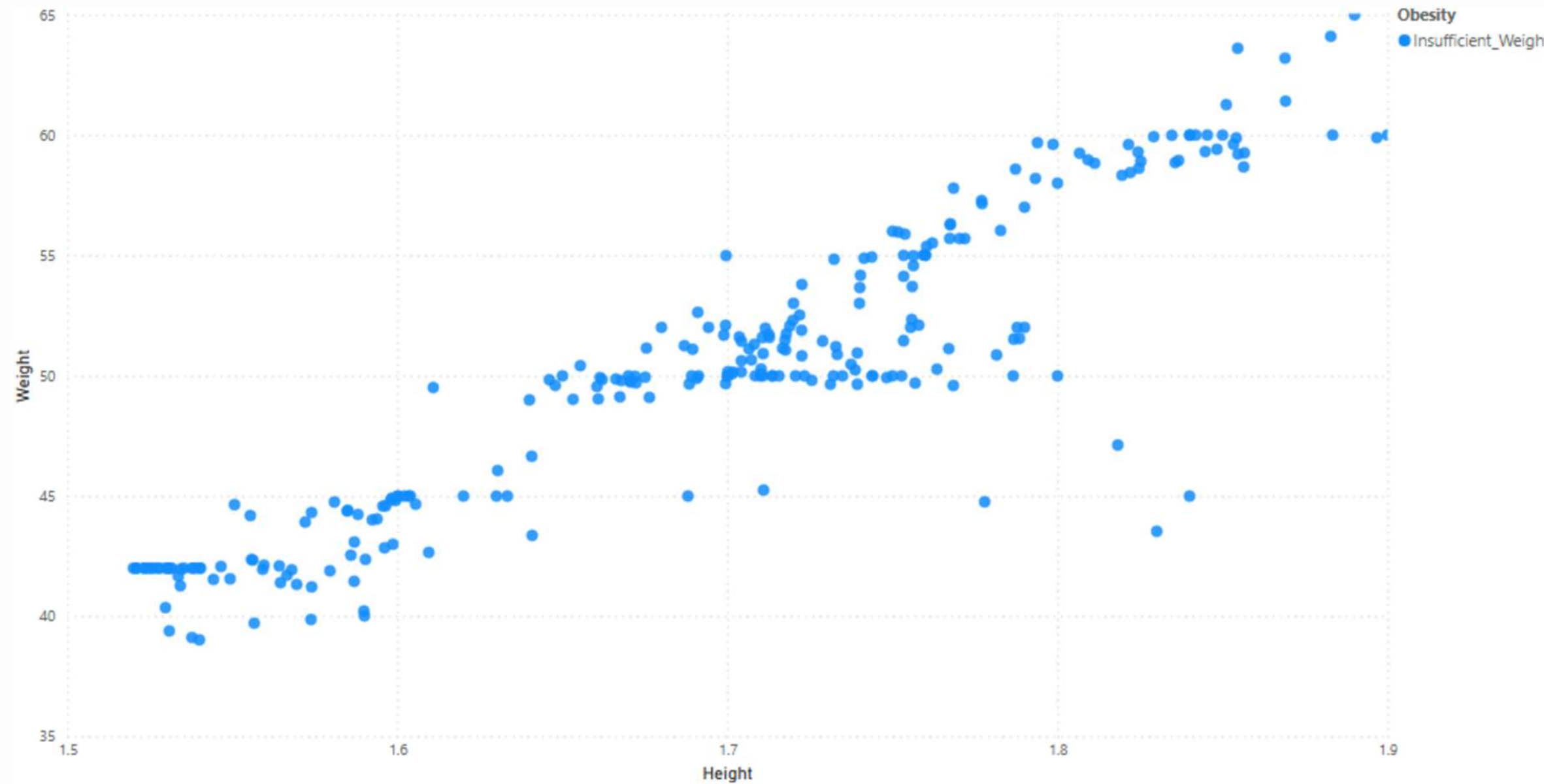
Linear Regression

8. Linear Regression



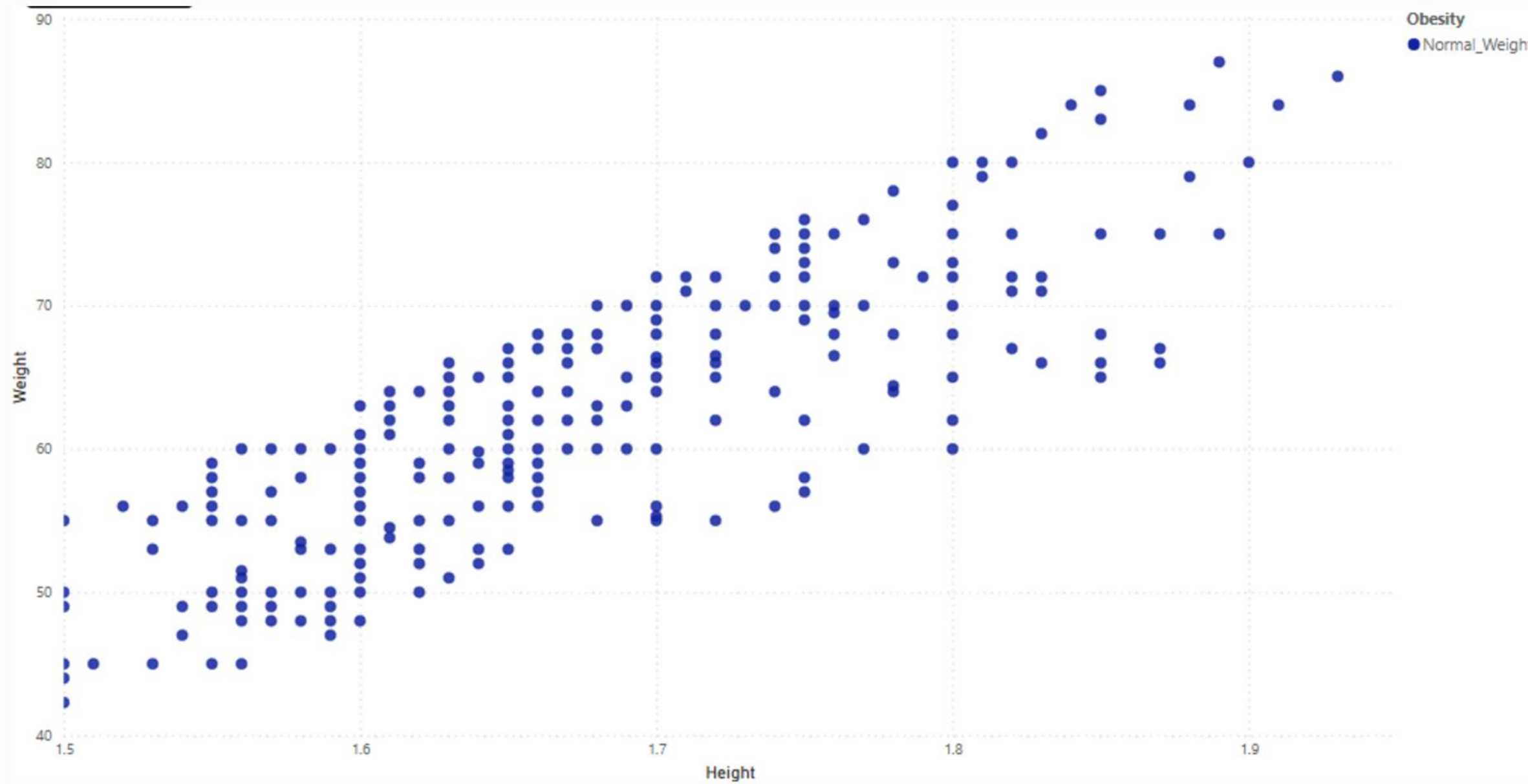
Linear Regression

8. Linear Regression



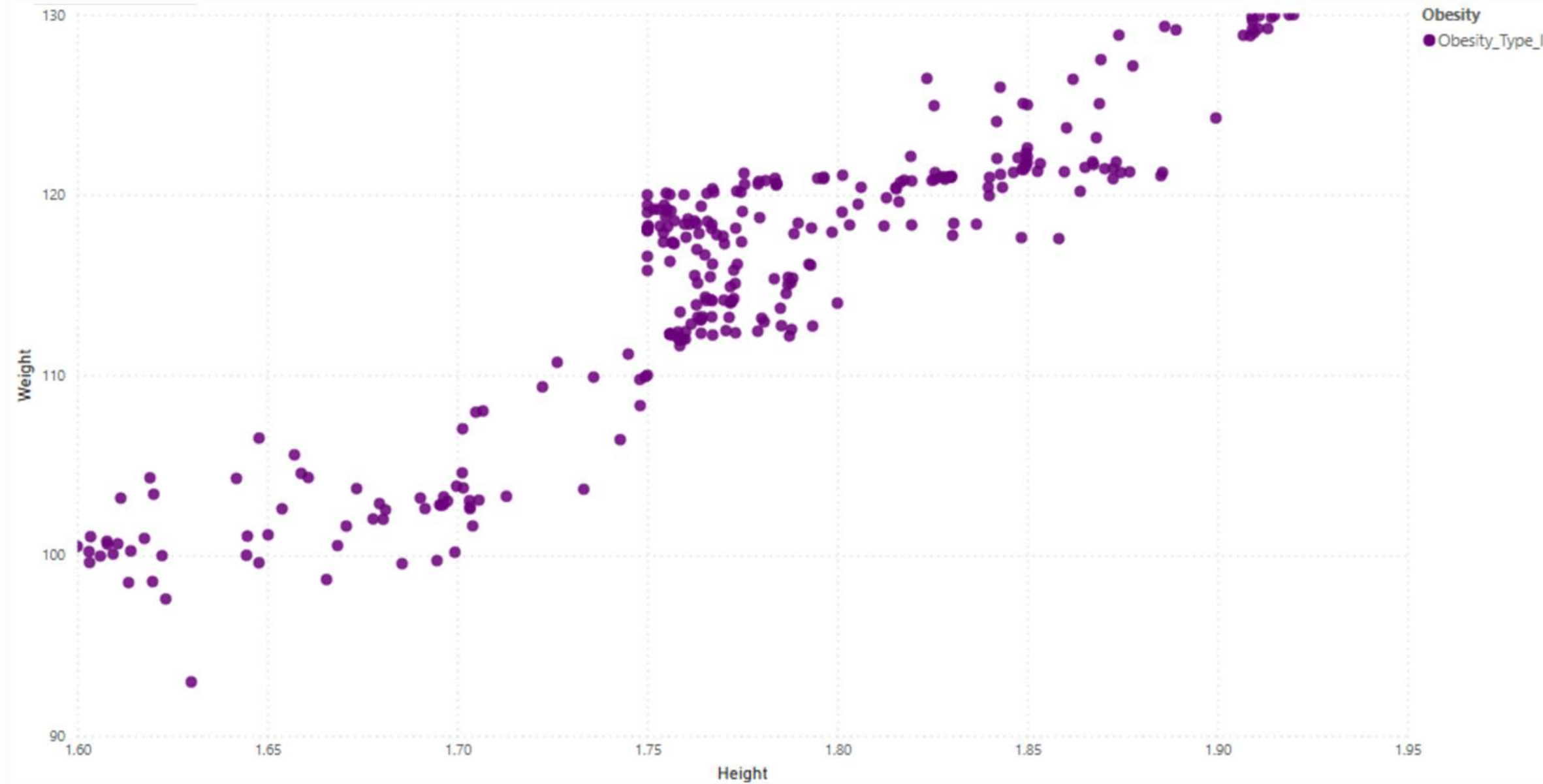
Linear Regression

8. Linear Regression



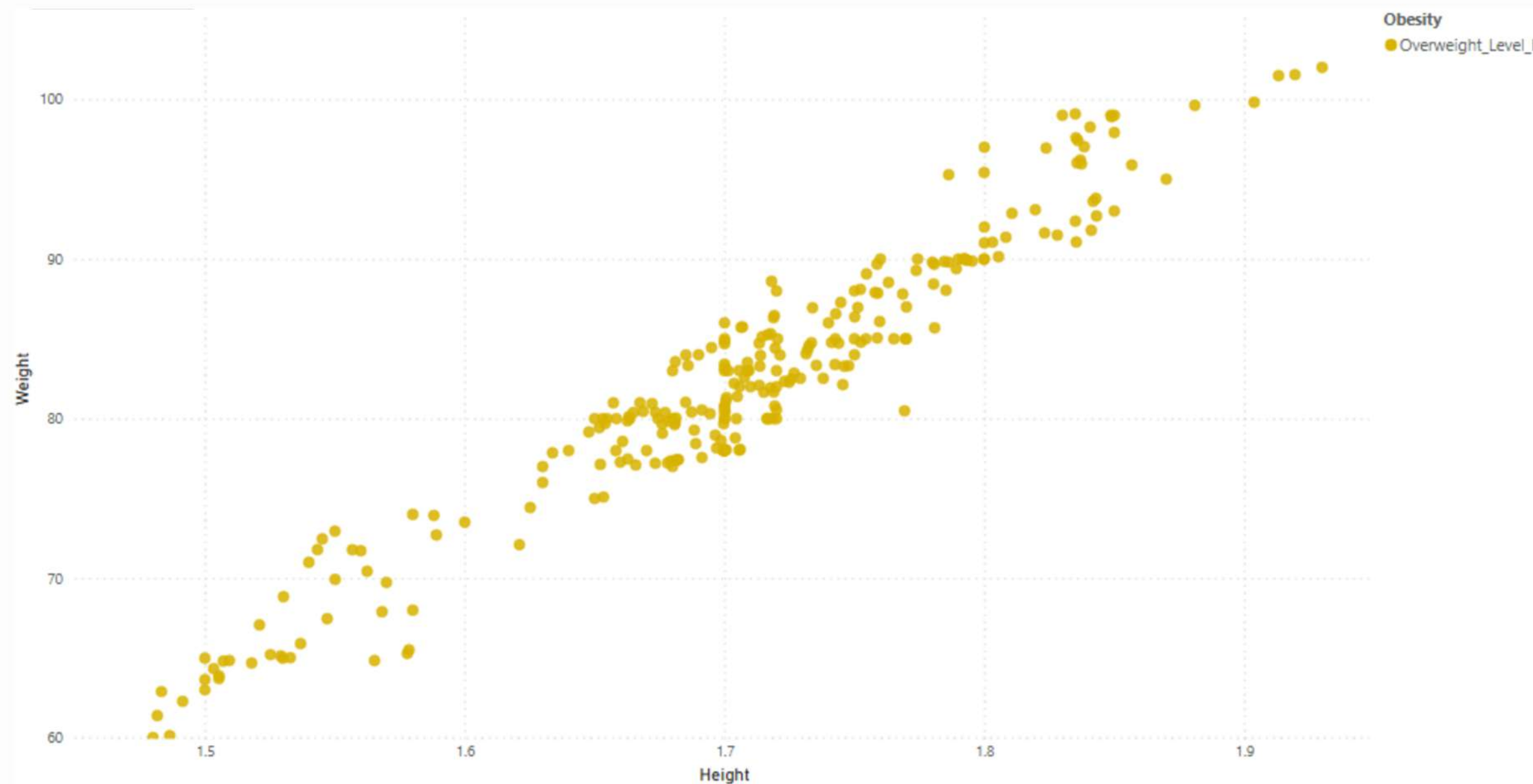
Linear Regression

8. Linear Regression



Linear Regression

8. Linear Regression



Linear Regression

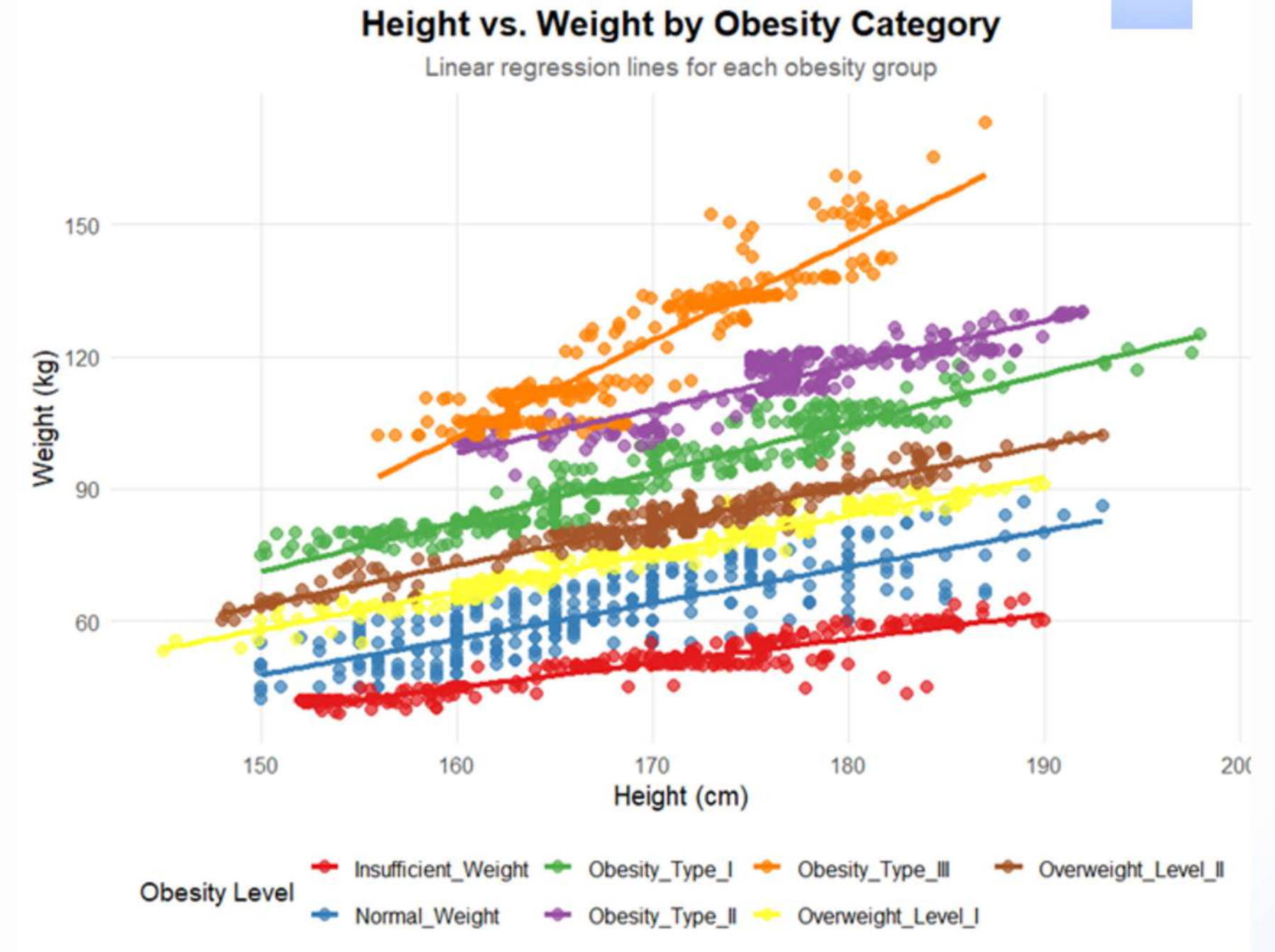
8. Linear Regression

Category	Intercept	Height Coef.	R ²	RMSE	p-value
Insufficient	-43.685	55.343	0.8425	2.39	< 2e-16
Normal	-74.634	81.588	0.6883	5.199	< 2e-16
Overweight I	-71.298	86.243	0.9553	1.795	< 2e-16
Overweight II	-72.512	90.740	0.9223	2.36	< 2e-16
Obesity I	-96.461	111.779	0.9173	3.308	< 2e-16
Obesity II	-62.166	100.165	0.8241	3.372	< 2e-16
Obesity III	-251.650	220.787	0.8606	5.808	< 2e-16

Linear Regression

8. Linear Regression

if we plot the height and the weight, we will get a kind of an upward trend no one from that that they are positively correlated so if one increases the other more likely to increase as shown in the following plot:



PCA

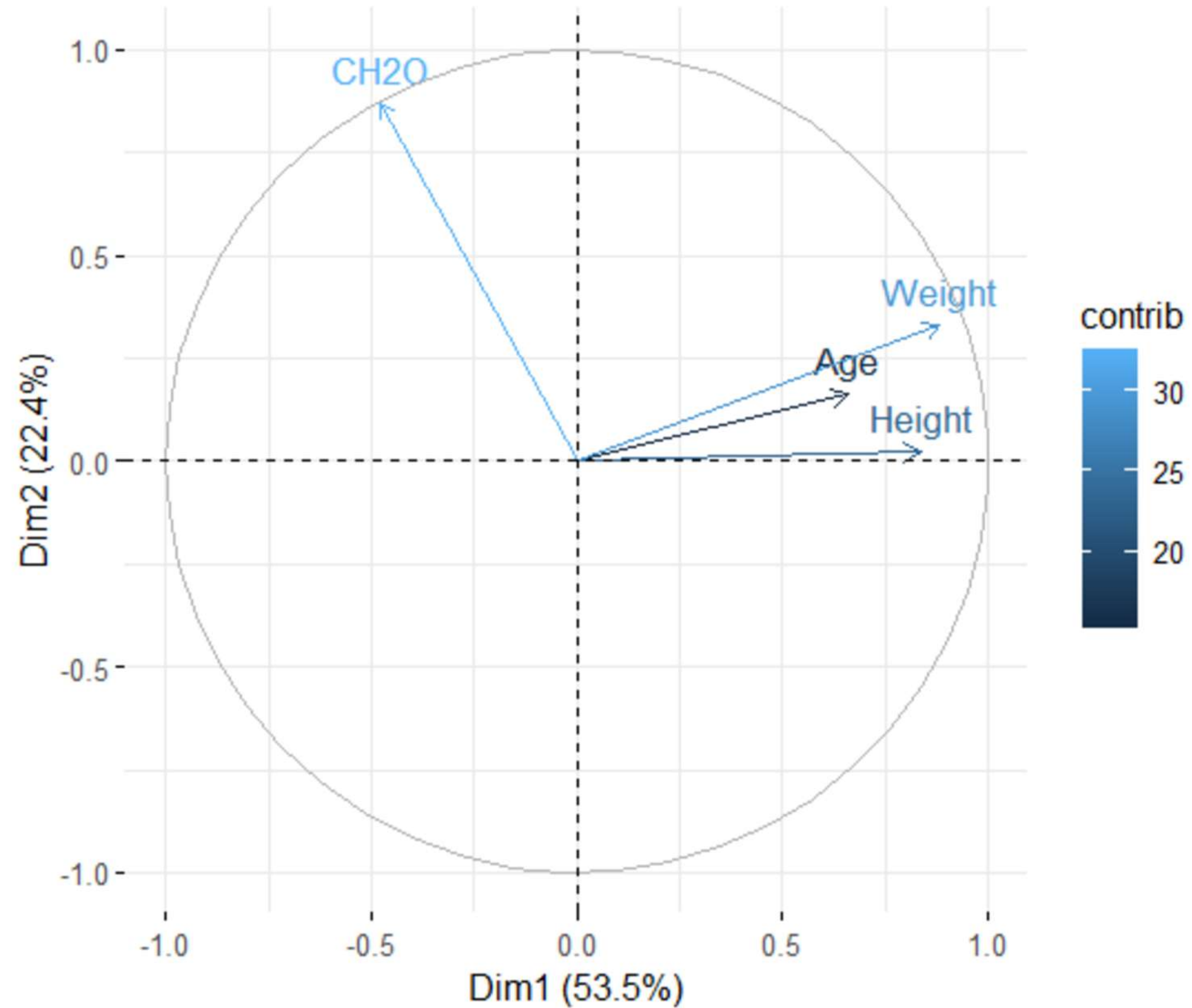
9. Principle Component Analysis (PCA)

Principal component analysis, or PCA, reduces the number of dimensions in large datasets to principal components that retain most of the original information. It does this by transforming potentially correlated variables into a smaller set of variables, called principal components.

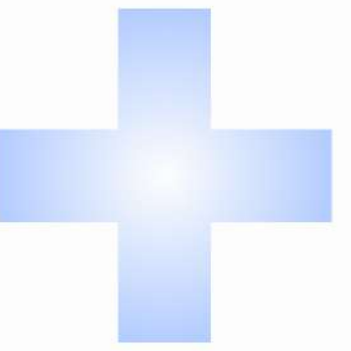
PCA

9. Principle Component Analysis (PCA)

Variables - PCA

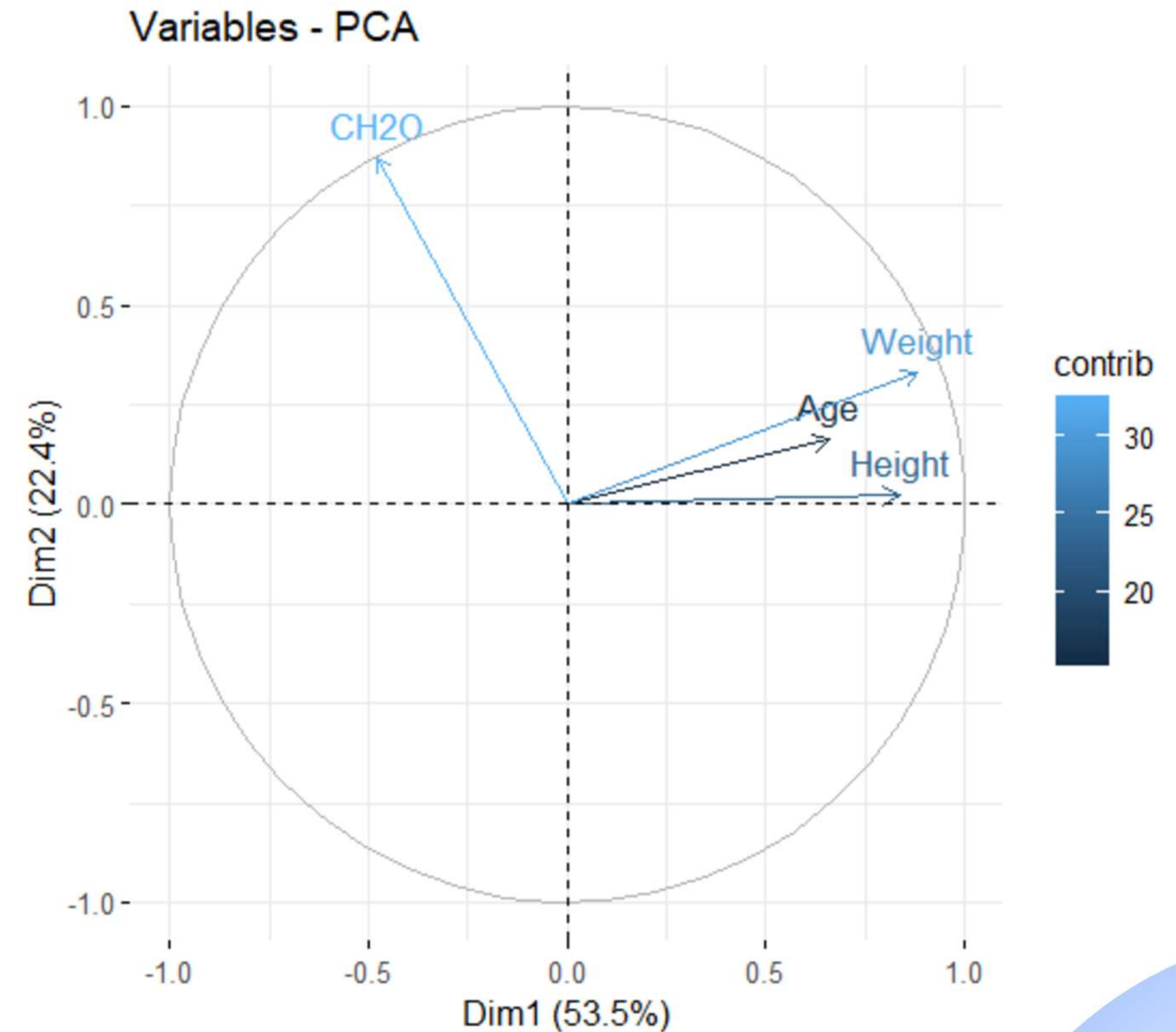


PCA

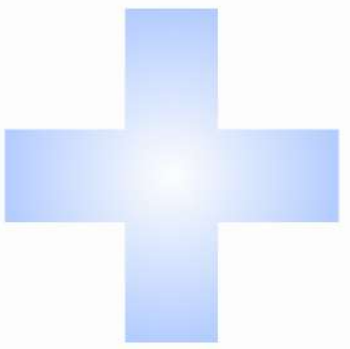


9. Principle Component Analysis (PCA)

- Dim1 seems to represent body size and age (Height, Weight, Age).
- Dim2 is more related to diet (CH2O intake).



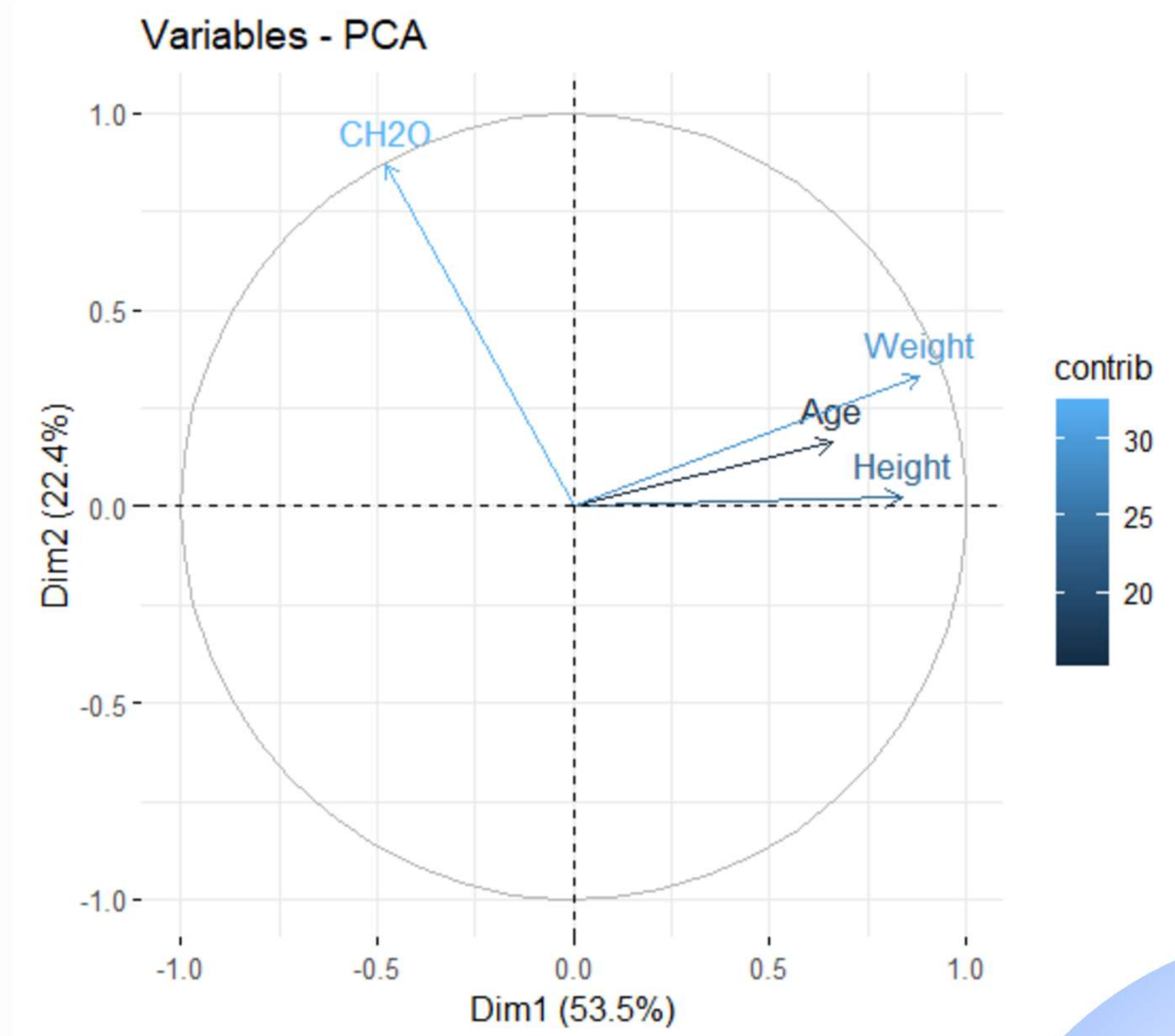
PCA



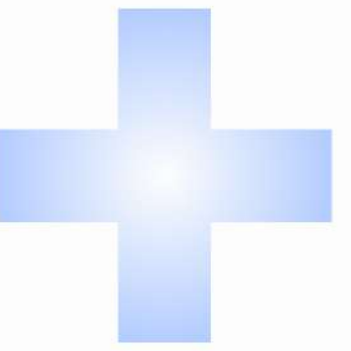
9. Principle Component Analysis (PCA)

Relationships Between Variables:

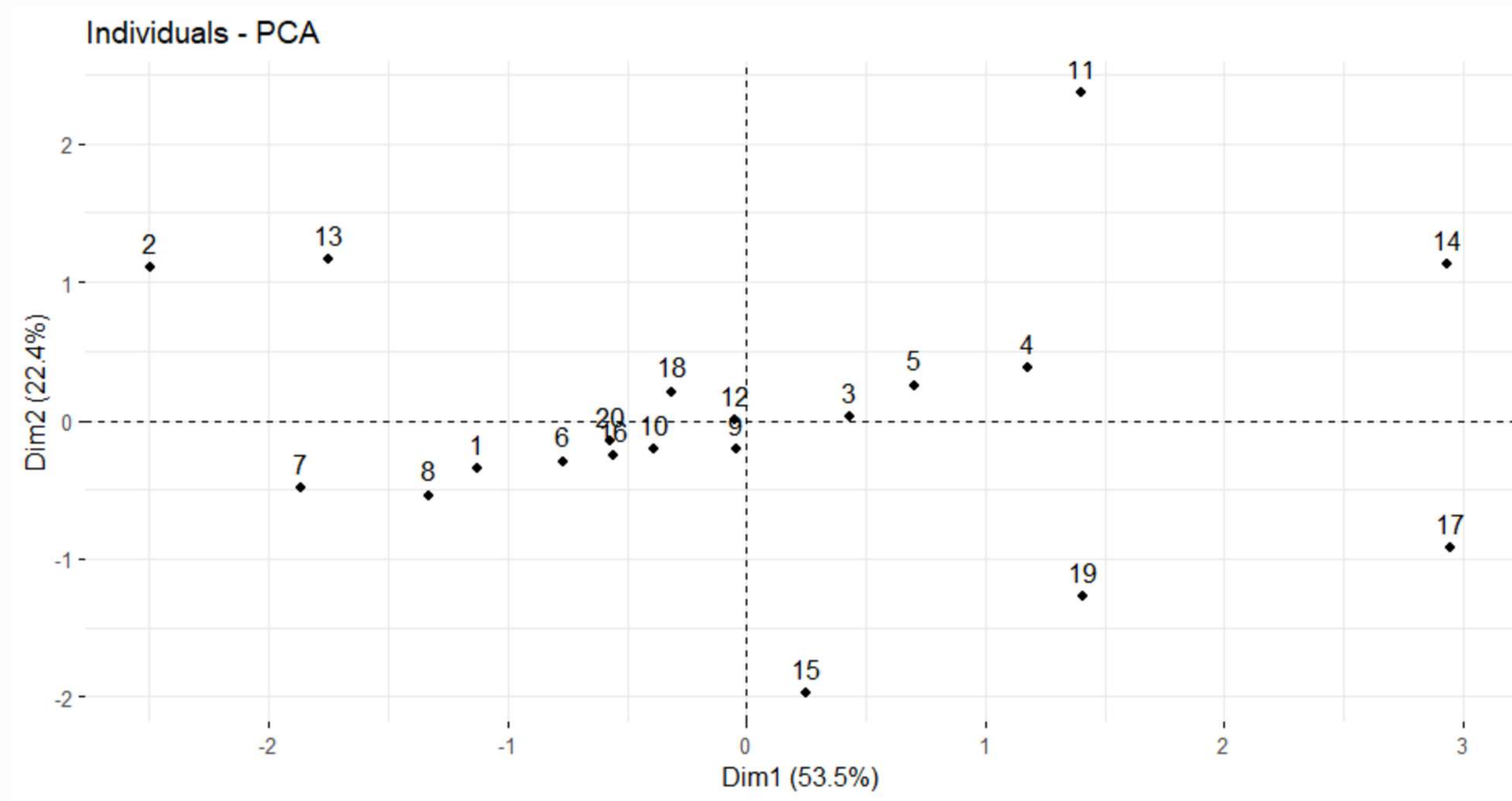
- Weight, Height, and Age are positively correlated with each other (same direction).
- CH2O is negatively correlated with them (points in the opposite direction). So, people who eat more carbs may tend to have lower weight or height, or belong to a different group (e.g., younger).



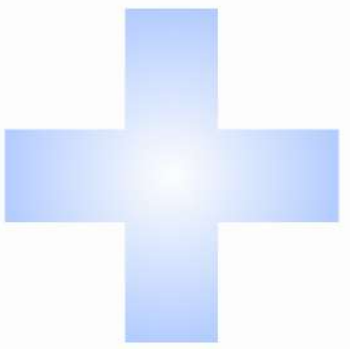
PCA



9. Principle Component Analysis (PCA)



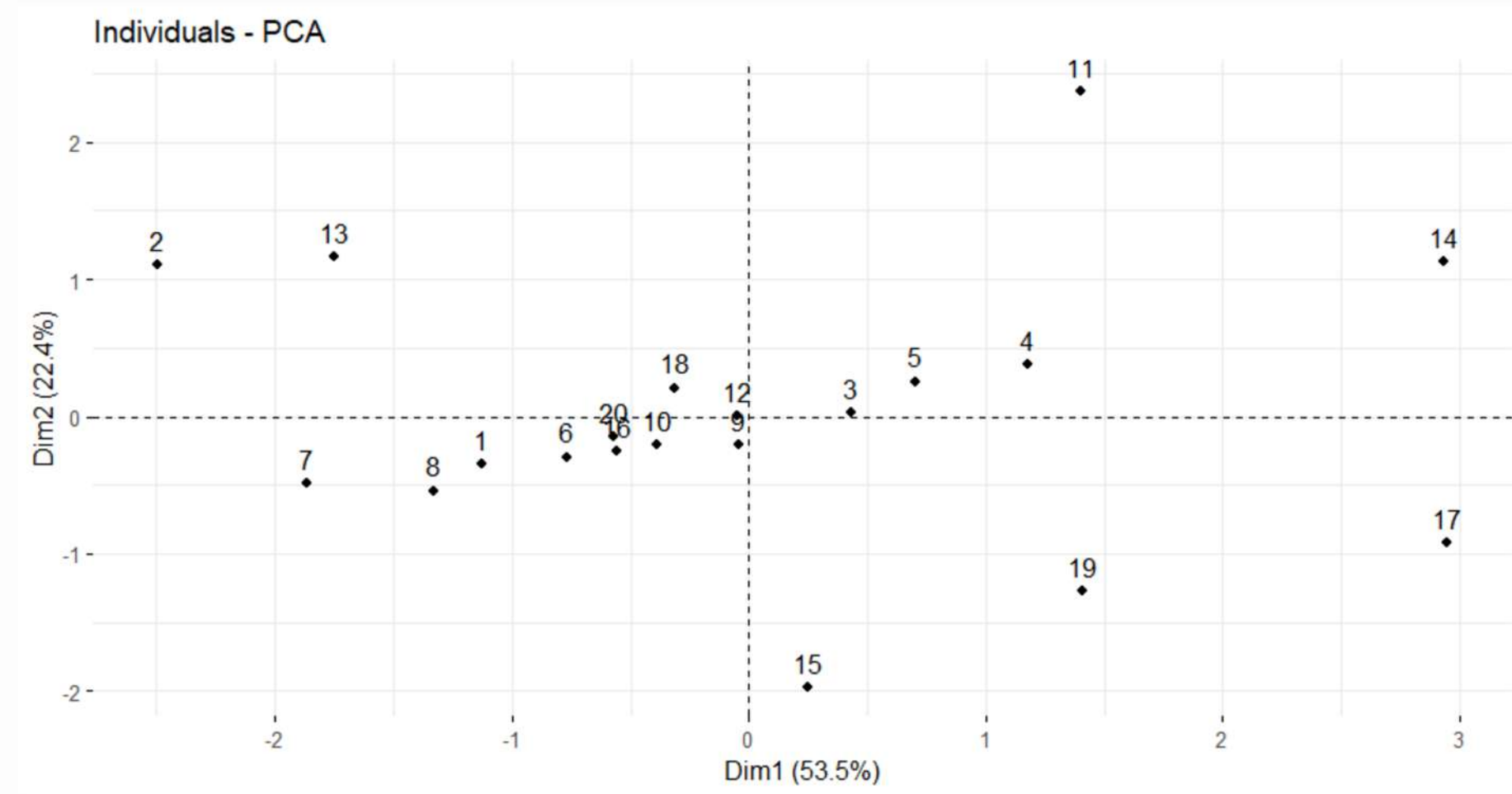
PCA



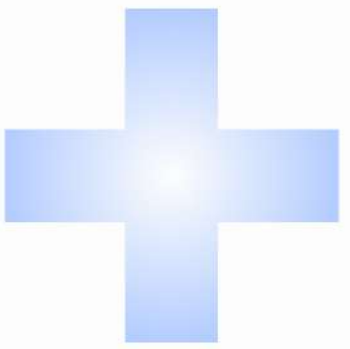
9. Principle Component Analysis (PCA)

Right side of Dim1 (positive values) — Likely high Height/Weight/Age:

- 14: Far right and slightly high → Very high Height/Weight, and moderate CH2O.
- 17: Far right and lower → Very high Height/Weight, possibly low CH2O.
- 11: High on both dimensions → High Height/Weight and CH2O.
- 4 & 5: Slightly high on both dimensions → Somewhat high in both Height/Weight and CH2O.
- 3: Near origin, slight positive Dim1 → Neutral to slightly above average values.



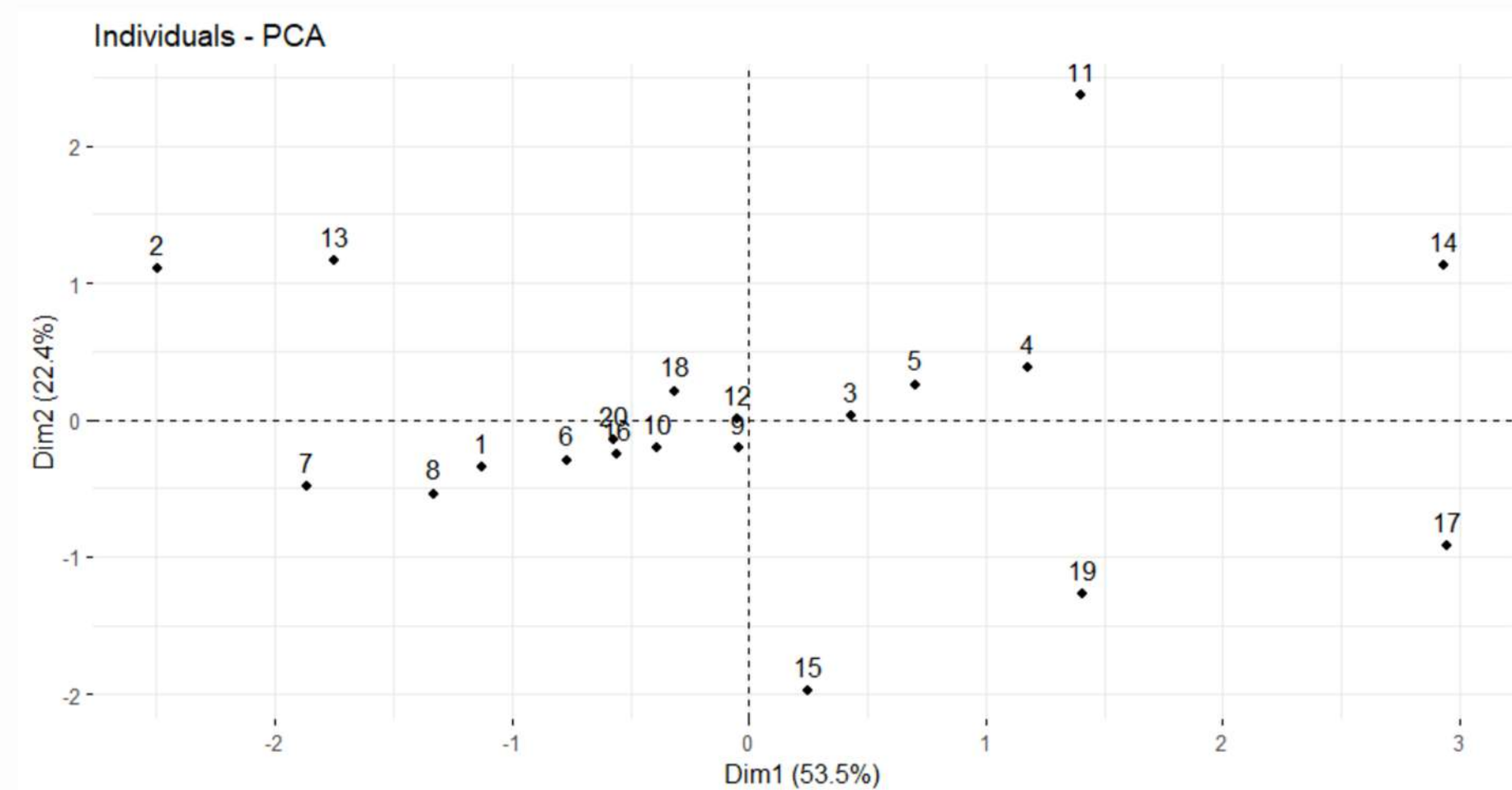
PCA



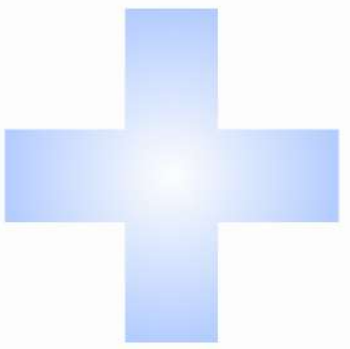
9. Principle Component Analysis (PCA)

Bottom of Dim2 (negative values) — Likely low CH₂O:

- 15: Most negative on Dim2 → Possibly very low CH₂O and average Height/Weight.
- 9: Close to center but slightly low → Near average, maybe a bit low on CH₂O.
- 18, 10, 12: Clustered near origin → Balanced or average across variables.



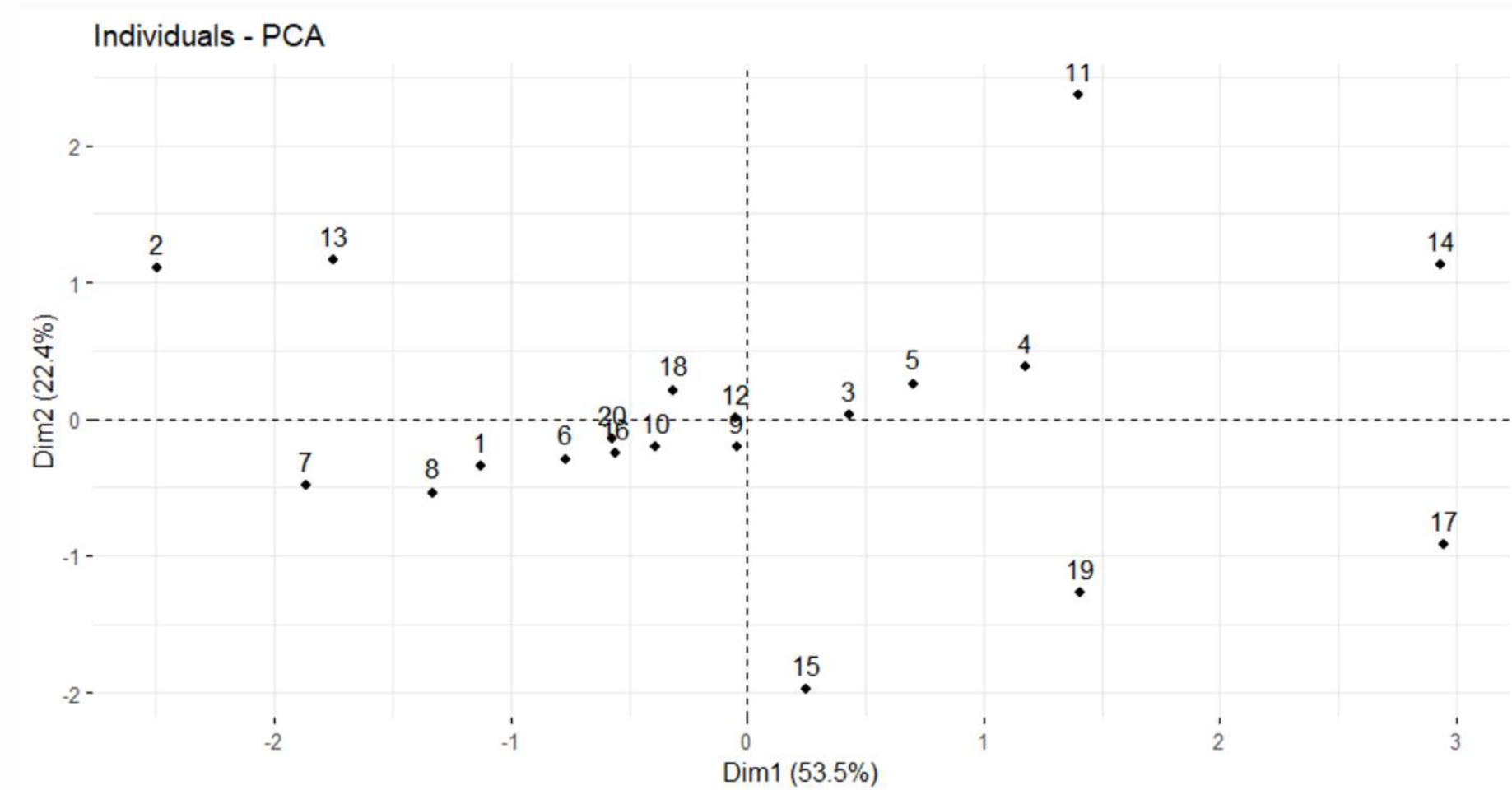
PCA



9. Principle Component Analysis (PCA)

Near the Origin (0,0) — Average individuals:

- 9, 10, 12, 16, 18, 20: No extreme values; these are your baseline or "typical" individuals.

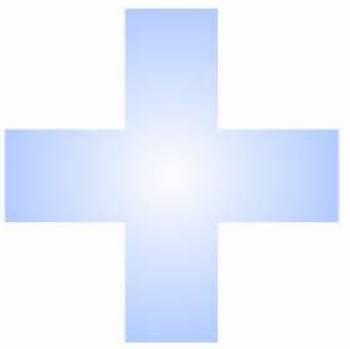


Conclusion



This report explored how daily habits affect obesity using the Obesity Level Dataset. The analysis showed that obesity is not random—it is strongly tied to how people live their everyday lives. First, most people in the dataset eat high-calorie food often, snack between meals, and don't monitor what they eat. Many also have a family history of being overweight, which could be due to genetics or shared habits at home. At the same time, physical activity is low. Very few walk or bike, while most use public transport or cars. All of these behaviors create a lifestyle where gaining weight becomes very likely. When we looked at the numbers, we saw that people who are obese tend to be older. Age seems to be an important factor. But height does not show any clear link to obesity. The ANOVA tests confirmed this: weight is very different across obesity levels, and height is also slightly different, but not as strongly. This means that obesity level clearly affects a person's weight, and to a lesser extent, their height. In short, the results confirm what many health experts say: poor eating habits, low physical activity, and certain family traits are the key reasons behind the rise in obesity. To prevent obesity, people need to eat better, move more, and develop healthy routines early in life. This study shows how simple daily choices can have big effects on health. Understanding these patterns can help design better health programs and raise awareness about the dangers of unhealthy lifestyles.

10. R Studio Scripts



1. Load Libraries and Dataset

```
> library(readxl)
> library(tidyverse)
> df <- read_excel("C:/Users/ULTRAPC/Desktop/Data Minig Project/group 4 Obesity Prediction Dataset.xlsx")
> View(df)
```

2. Summary Statistics

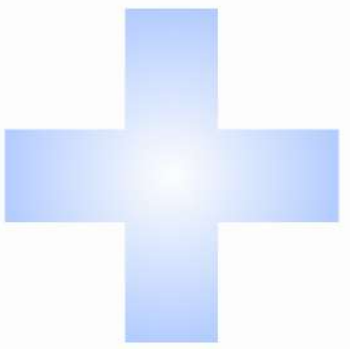
Numeric Variables

```
> numeric_data <- df[apply(df, is.numeric)]
> summary(numeric_data)
```

Categorical Variables

```
> categorical_data <- df[apply(df, function(x) is.factor(x) | is.character(x))]
> summary(categorical_data)
```

10. R Studio Scripts



3. Empirical Analysis for Numeric Variables:

```
> mean(df$Age)
> median(df$Age)
> sd(df$Age)
> quantile(df$Age)
> min(df$Age)
> max(df$Age)

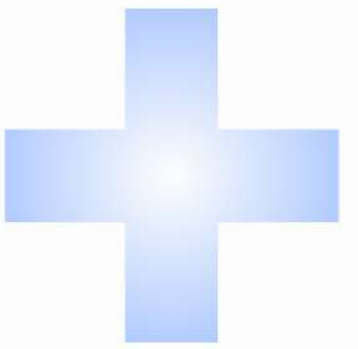
> mean(df$Weight)
> median(df$Weight)
> sd(df$Weight)
> quantile(df$Weight)
> min(df$Weight)
> max(df$Weight)

> df$Height_cm <- df$Height * 100
> mean(df$Height_cm)
> median(df$Height_cm)
> sd(df$Height_cm)
> quantile(df$Height_cm)
> min(df$Height_cm)
> max(df$Height_cm)
```

4. Empirical Analysis for Categorical Variables

```
> table(df$Gender)
> table(df$family_history)
> table(df$FAVC)
> table(df$CAEC)
> table(df$SMOKE)
> table(df$SCC)
> table(df$CALC)
> table(df$MTRANS)
> table(df$Obesity)
```

10. R Studio Scripts



5. Charts for Categorical Variables

Gender Distribution

```
> ggplot(df, aes(x = Gender)) +  
  geom_bar(fill = "steelblue", color = "black") +  
  labs(title = "Gender Distribution", x = "Gender", y = "Count") +  
  theme_minimal()
```

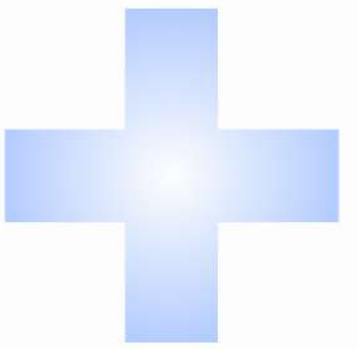
CAEC Distribution

```
> ggplot(df, aes(x = CAEC)) +  
  geom_bar(fill = "steelblue", color = "black") +  
  labs(title = "Eating Habit (CAEC)", x = "CAEC", y = "Count") +  
  theme_minimal()
```

MTRANS Distribution

```
> ggplot(df, aes(x = MTRANS)) +  
  geom_bar(fill = "steelblue", color = "black") +  
  labs(title = "Transportation Mode (MTRANS)", x = "MTRANS", y = "Count") +  
  theme_minimal()
```

10. R Studio Scripts



6. Charts for Numeric Variables

Age

```
> ggplot(df, aes(x = Age)) +  
  geom_histogram(aes(y = after_stat(density)), binwidth = 5, fill = "steelblue", color = "white", alpha = 0.7) +  
  geom_density(color = "darkblue", linewidth = 1.2) +  
  labs(title = "Age Distribution", x = "Age", y = "Density") +  
  theme_minimal()
```

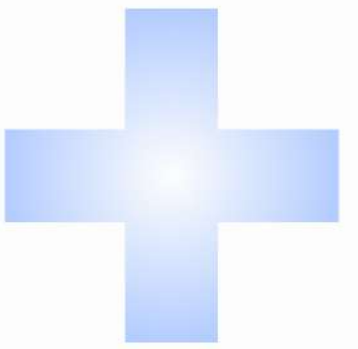
Height (in cm)

```
> ggplot(df, aes(x = Height_cm)) +  
  geom_histogram(aes(y = after_stat(density)), binwidth = 5, fill = "steelblue", color = "white", alpha = 0.7) +  
  geom_density(color = "darkblue", linewidth = 1.2) +  
  labs(title = "Height Distribution", x = "Height (cm)", y = "Density") +  
  theme_minimal()
```

Weight

```
> ggplot(df, aes(x = Weight)) +  
  geom_histogram(aes(y = after_stat(density)), binwidth = 5, fill = "steelblue", color = "white", alpha = 0.7) +  
  geom_density(color = "darkblue", linewidth = 1.2) +  
  labs(title = "Weight Distribution", x = "Weight (kg)", y = "Density") +  
  theme_minimal()
```

10. R Studio Scripts



8. ANOVA Test: Numeric vs Obesity

Weight ~ Obesity

```
> aov_weight <- aov(Weight ~ Obesity, data = df)
```

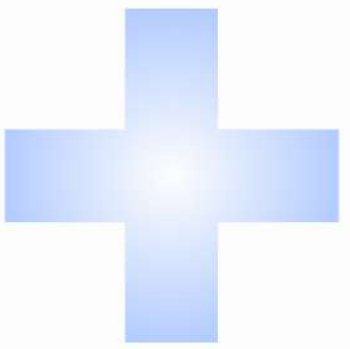
```
> summary(aov_weight)
```

Height ~ Obesity

```
> aov_height <- aov(Height ~ Obesity, data = df)
```

```
> summary(aov_height)
```

10. R Studio Scripts



9. Scatter Plots and Linear Regression by Obesity Type

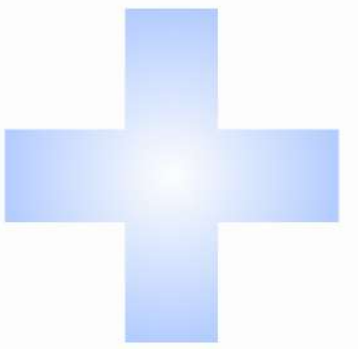
Scatter Plot: Height vs Weight

```
> ggplot(data = df, aes(x = Height, y = Weight)) +  
  geom_point(  
    size = 3,  
    shape = 21,  
    color = "black",  
    fill = "steelblue"  
  ) +  
  labs(  
    title = "Relationship Between Height and Weight",  
    subtitle = "A scatter plot showing the correlation between individuals' height and weight",  
    x = "Height (m)",  
    y = "Weight (kg)",  
  ) +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),  
    plot.subtitle = element_text(size = 12, hjust = 0.5, color = "grey40")  
  )
```

Scatter Plot with Obesity Categories

```
> ggplot(data = df, aes(x = Height, y = Weight, color = Obesity)) +  
  geom_point(size = 2.5, alpha = 0.8) + # No border, smaller points  
  scale_color_brewer(palette = "Set1") + # Colorblind-friendly palette  
  labs(  
    x = "Height",  
    y = "Weight",  
    color = "Obesity" # Legend title  
  ) +  
  theme_minimal() +  
  theme(  
    legend.position = "right", # Legend on the side  
    plot.title = element_blank(),  
    plot.subtitle = element_blank()  
  )
```

10. R Studio Scripts



10. Perform Linear Regression by Obesity Type

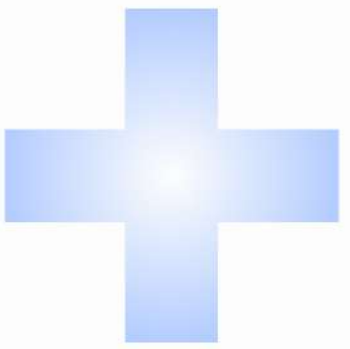
Linear Regression by Obesity Type

```
> # Create a list to store regression results
> regression_results <- list()

> # Perform linear regression for each obesity type
> for (type in obesity_types) {
> temp_df <- obesity_dfs[[type]]
> if (nrow(temp_df) > 1) { # Need at least 2 points for regression
>   model <- lm(Weight ~ Height_cm, data = temp_df)
>   regression_results[[type]] <- summary(model)
> } else {
>   regression_results[[type]] <- paste("Not enough data for", type)
> }
> }

> # Print all results
> for (type in names(regression_results)) {
> cat("\n=== Regression results for", type, "===\n")
> print(regression_results[[type]])
> }
```

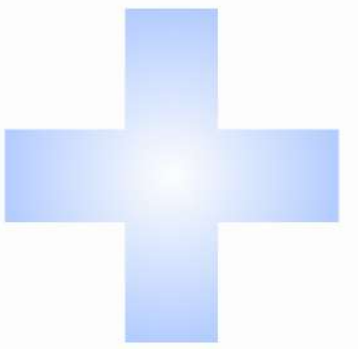
10. R Studio Scripts



11. Linear Regression with Smooth Curves

```
> ggplot(df, aes(x = Height_cm, y = Weight, color = Obesity)) +  
  geom_point(size = 3, alpha = 0.7) + # Add transparency to help with overlapping  
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.3) + # Thicker line for clarity  
  scale_color_brewer(palette = "Set1") + # Colorblind-friendly palette  
  labs(  
    title = "Height vs. Weight by Obesity Category",  
    subtitle = "Linear regression lines for each obesity group",  
    x = "Height (cm)",  
    y = "Weight (kg)",  
    color = "Obesity Level"  
  ) +  
  theme_minimal(base_size = 14) + # Larger base font for better readability  
  theme(  
    plot.title = element_text(size = 18, face = "bold", hjust = 0.5),  
    plot.subtitle = element_text(size = 13, hjust = 0.5, color = "gray40"),  
    legend.position = "bottom",  
    legend.title = element_text(size = 13),  
    legend.text = element_text(size = 11),  
    panel.grid.minor = element_blank(), # Cleaner background  
    panel.grid.major = element_line(color = "gray90")  
  )
```

10. R Studio Scripts



12. Principle Component Analysis (PCA)

```
library(FactoMineR)  
library(factoextra)
```

```
pca_result <- prcomp(num_df, scale. = TRUE)
```

```
screeplot(pca_result, type = "lines")  
fviz_eig(pca_result)  
fviz_pca_var(pca_result, col.var = "contrib",)  
fviz_pca_ind(pca_result)
```